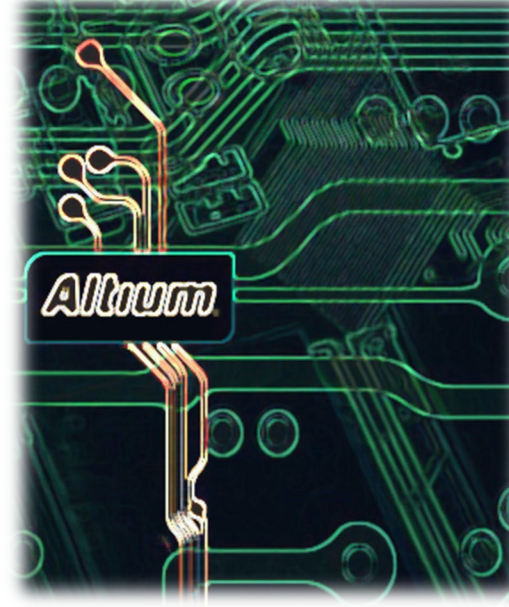


ALTIUM DESIGNER

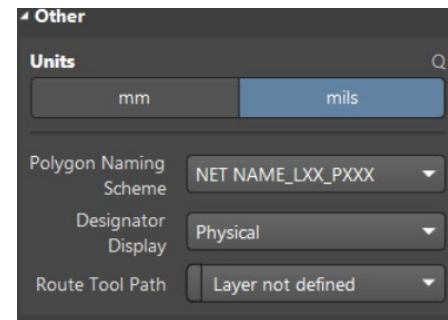
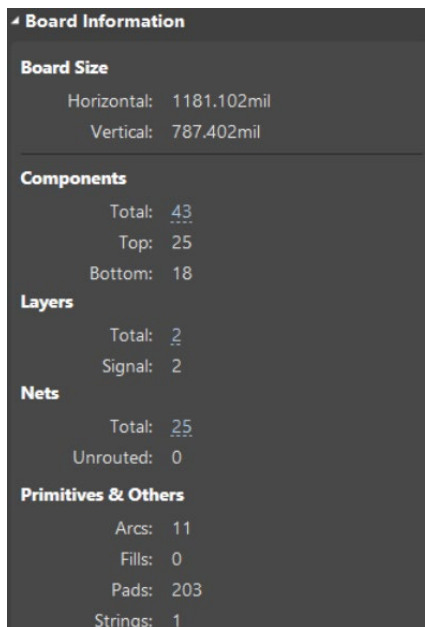
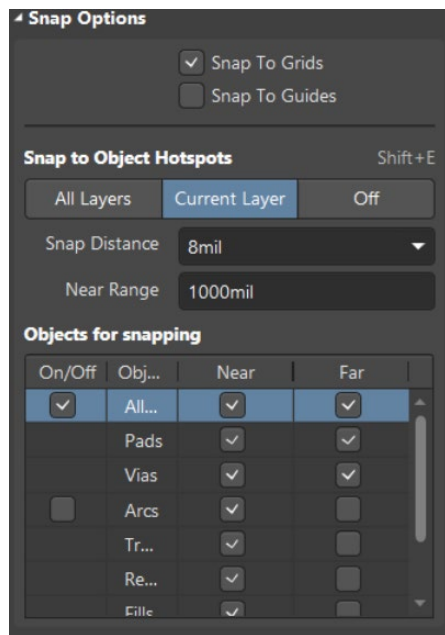
DISEÑO DE LAYOUT



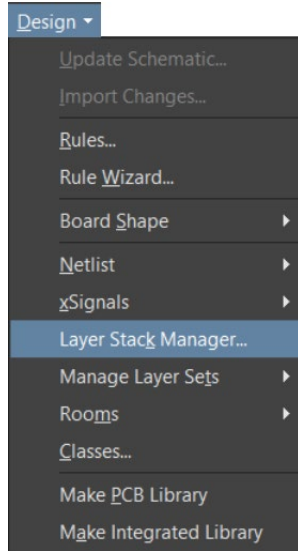
Basado en materiales elaborados para Electratraining (www.electratraining.org) por:
Miguel Albero Gil (malbero@ai2.upv.es)

BOARD OPTIONS

From Properties Panel (you must deselect any element), and a with a PCB document opened, you can define all work document in a single panel.



LAYERS DEFINITION



Board Information	
Board Size	
Horizontal:	1181.102mil
Vertical:	787.402mil
Components	
Total:	43
Top:	25
Bottom:	18
Layers	
Total:	2
Signal:	2
Nets	
Total:	25
Unrouted:	0
Primitives & Others	
Arcs:	11
Fills:	0
Pads:	203
Strings:	1

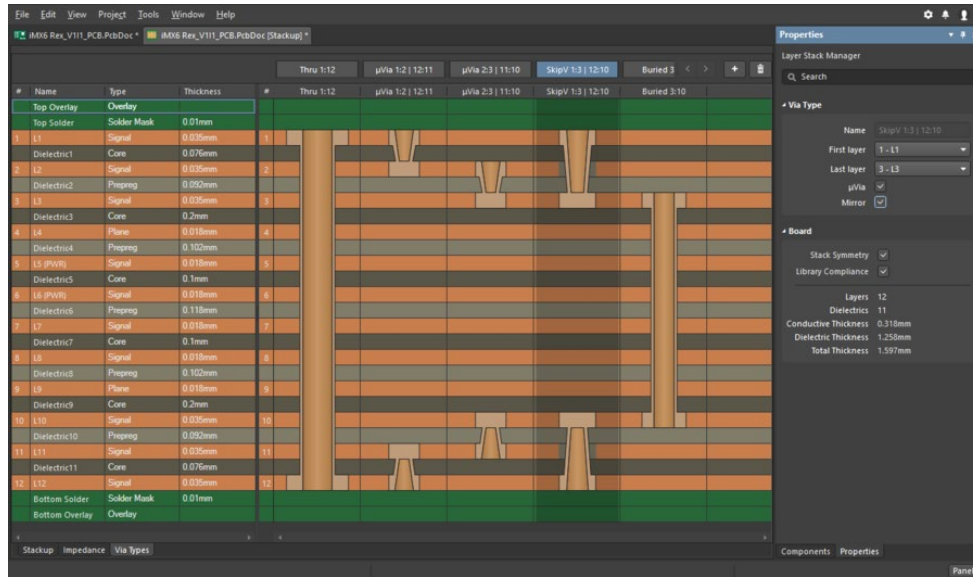
All layer stacks are defined in the *Layer Stack Manager*. To open the *Layer Stack Manager* select **Design » Layer Stack Manager** from the menus, or from a Properties panel, clicking over number of layers. For a new board, its single default stack comprises: a dielectric core, 2 copper layers, as well as the top and bottom solder/overlay and overlay layers.

The screenshot shows the 'Layer Stack Manager' dialog with a table of 16 layers. An orange arrow points from the 'Layers' section of the 'Board Information' panel to the table.

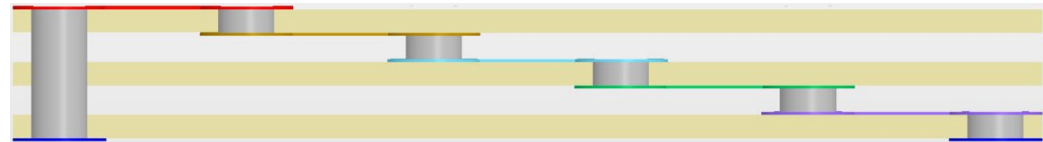
#	Name	Material	Type	Thickness	Weight	Dr	Df	Process
1	Top Overlay	Solder Resist	Overlay	0.01mm				
2	Top Solder	Solder Mask			3.5			
3	1_top	CF-004	Signal	0.050mm	1oz			ED
4	Dielectric 1	TG-883	Core	0.1mm	3.6			ED
5	2_int1_signal	CF-003	Signal	0.018mm	15oz			ED
6	Dielectric 2	TG-883	Prepreg	0.1mm	3.6			ED
7	3_int1_prepeg	CF-003	Signal	0.018mm	15oz			ED
8	Dielectric 3	TG-883	Core	0.1mm	3.6			ED
9	4_int1_signal	CF-003	Signal	0.018mm	15oz			ED
10	Dielectric 4	TG-883	Prepreg	0.12mm	3.6			ED
11	5_int1_signal	CF-003	Signal	0.018mm	15oz			ED
12	Dielectric 5	TG-883	Core	0.12mm	3.6			ED
13	6_int1_prepeg	CF-003	Signal	0.018mm	15oz			ED
14	Dielectric 6	TG-883	Prepreg	0.12mm	3.6			ED
15	7_int1_signal	CF-003	Signal	0.018mm	15oz			ED
16	Dielectric 7	TG-883	Core	0.12mm	3.6			ED
17	8_int1_prepeg	CF-003	Signal	0.018mm	15oz			ED
18	Dielectric 8	TG-883	Prepreg	0.12mm	3.6			ED
19	9_int1_signal	CF-003	Signal	0.018mm	15oz			ED
20	Dielectric 9	TG-883	Core	0.12mm	3.6			ED
21	10_int1_prepeg	CF-003	Signal	0.018mm	15oz			ED
22	Dielectric 10	TG-883	Prepreg	0.12mm	3.6			ED
23	11_int12_signal	CF-003	Signal	0.018mm	15oz			ED
24	Dielectric 11	TG-883	Core	0.12mm	3.6			ED
25	12_int11_topgt	CF-003	Signal	0.018mm	15oz			ED
26	Dielectric 12	TG-883	Prepreg	0.12mm	3.6			ED
27	13_int12_signal	CF-003	Signal	0.018mm	15oz			ED
28	Dielectric 13	TG-883	Core	0.1mm	3.6			ED
29	14_int13_prepeg	CF-003	Signal	0.018mm	15oz			ED
30	Dielectric 14	TG-883	Prepreg	0.1mm	3.6			ED
31	15_int12_signal	CF-003	Signal	0.018mm	15oz			ED
32	Dielectric 15	TG-883	Core	0.1mm	3.6			ED
33	16_bottom	CF-004	Signal	0.050mm	1oz			ED
34	Bottom Solder	Solder Resist	Overlay	0.01mm	3.5			
35	Bottom Overlay	Solder Mask	Overlay					

Properties	
Layer Stack Manager	
Q Search	
Layer	
Name	12_int11_signal
Manufacturer	Altium Designer
Material	CF-003
Process	ED
Weight	0.5oz
Thickness	0.018mm
Orientation	Not allowed
Description	Copper Foil
Note	
Comment	
Board	
Stack Symmetry	☒
Library Compliance	☒
Layers 16	
Dielectrics	15
Conductive Thickness	0.12mm
Dielectric Thickness	1.6mm
Total Thickness	2.02mm

LAYERS DEFINITION

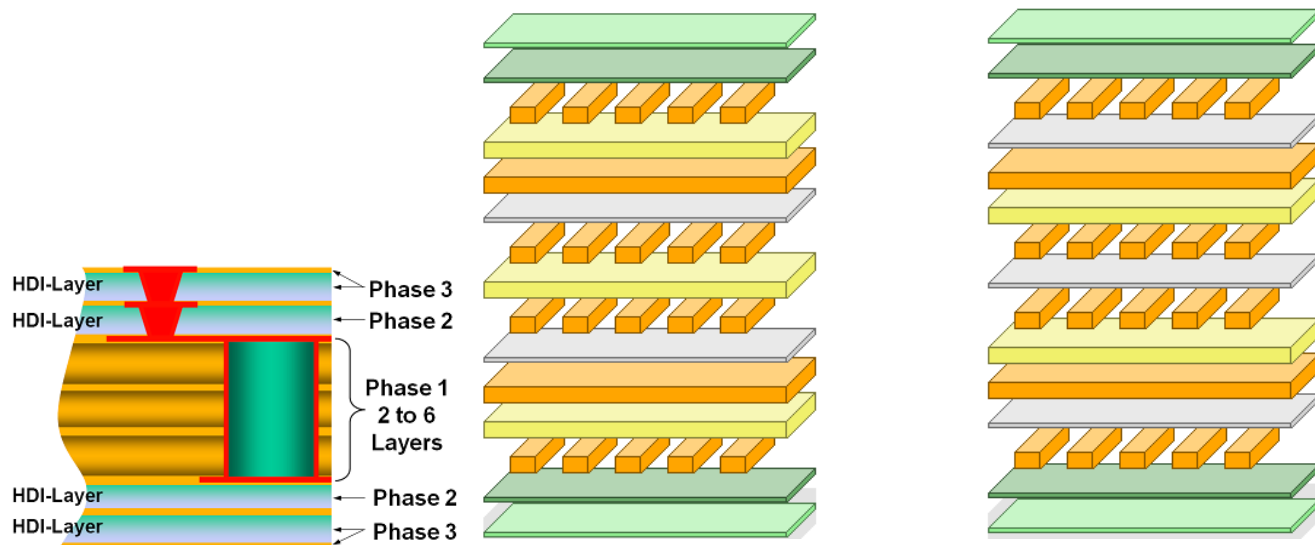


We should define the different layer drill pairs, based on our design needs and stackup definition from our PCB manufacturer, selecting Via Types



LAYERS DEFINITION

We can define our build-up layer distribution, depending on our PCB manufacturer



LAYERS DEFINITION

To fully document the layer stack, place a Layer Stack Table (**Place » Layer Stack Table**).
The Layer Stack Table details the layers, material, thickness, and dielectric constant.

Symbol	Count	Hole Size	Plated	Hole Type	Drill Layer Pair	Uia/Pad	Pad Shape	Template	Hole Tolerance (+)
⌀	1	25.00mil (0.635mm)	NPTH	Round	Top Layer - Bottom Layer	Pad	Rounded	c99r6B	
⌀	1	35.37mil (1.000mm)	PTH	Slot	Top Layer - Bottom Layer	Pad	Rounded	r200_c100h100_350-100	
⌀	2	25.37mil (1.000mm)	PTH	Slot	Top Layer - Bottom Layer	Pad	Rounded	(Fixed)	
⌀	3	43.31mil (1.100mm)	PTH	Round	Top Layer - Bottom Layer	Pad	Rounded	c330h110	
⌀	4	86.43mil (2.200mm)	NPTH	Round	Top Layer - Bottom Layer	Pad	Rounded	c230r0280	
⌀	4	125.82mil (3.200mm)	NPTH	Round	Top Layer - Bottom Layer	Pad	Rounded	(Fixed)	
⌀	4	25.83mil (0.790mm)	PTH	Round	Top Layer - Bottom Layer	Pad	(Fixed)	(Fixed)	
⌀	98	31.80mil (0.800mm)	PTH	Round	Top Layer - Bottom Layer	(Fixed)	Rounded	(Fixed)	
⌀	98	35.43mil (0.900mm)	PTH	Round	Top Layer - Bottom Layer	Pad	(Fixed)	(Fixed)	
⌀	170	23.42mil (0.600mm)	PTH	Round	Top Layer - Bottom Layer	(Fixed)	Rounded	(Fixed)	
944 Total									

Slot definitions: Routed Path Length = Calculated from tool start centre position to tool end centre position.
Hole Length = Routed Path Length + Tool Size = Slot length as defined in the PCB layout

Properties

Layer Stack Table

Components (and 12 more)

Search

Location

(X/Y)

0mm35.433mm

Rotation

0.000

Properties

Layer

Top Layer

Units

Imperial (mil)

Show Board Map

Line Width

0.381mm

Text Height

0.254mm

Font Type

TrueType

Stroke

Font

Default

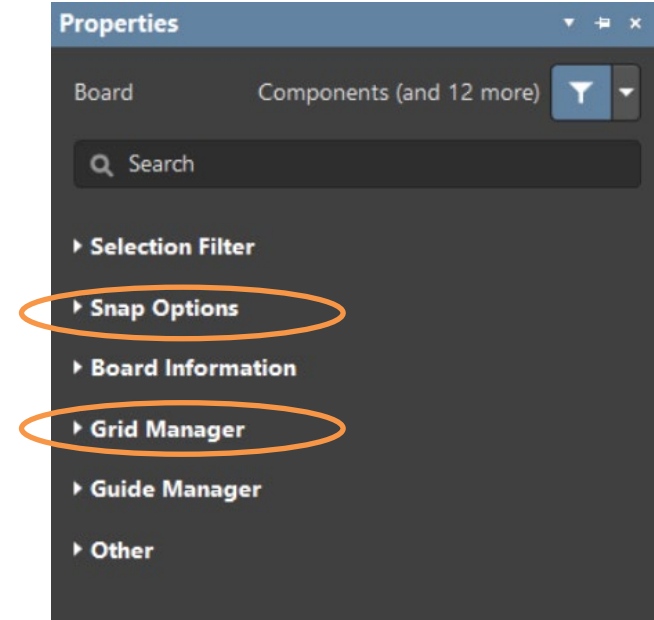
Stroke Width

0.254mm

1 object is displayed

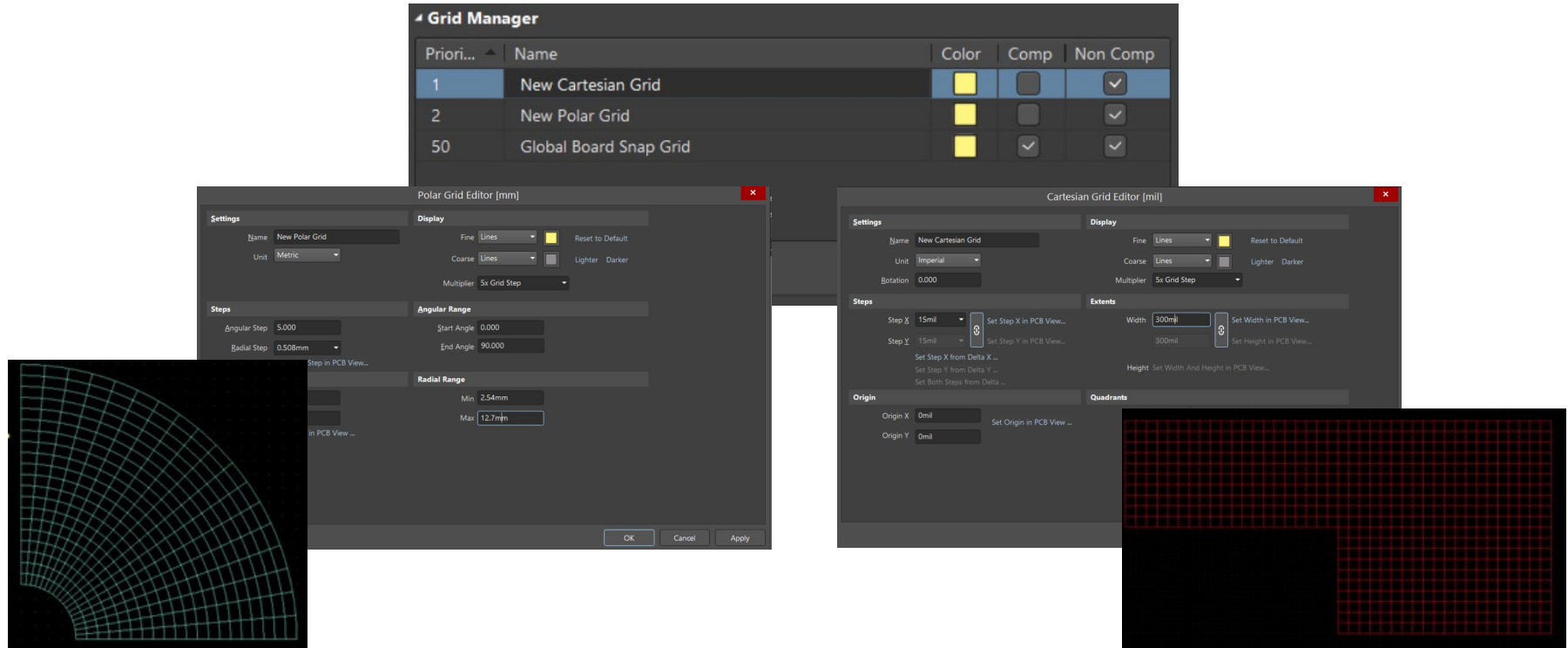
GRIDS

- This system brings together three different sub-systems to collectively drive the way that the cursor snaps:
 - *User-Definable Grids*, available in both Cartesian and Polar flavors;
 - *Snap Guides*, that can be freely placed and provide a handy visual cue for object alignment; and enhanced object
 - *Snap-Points*, enabling placed objects to pull the cursor into position based on cursor proximity to an object's hotspot(s)



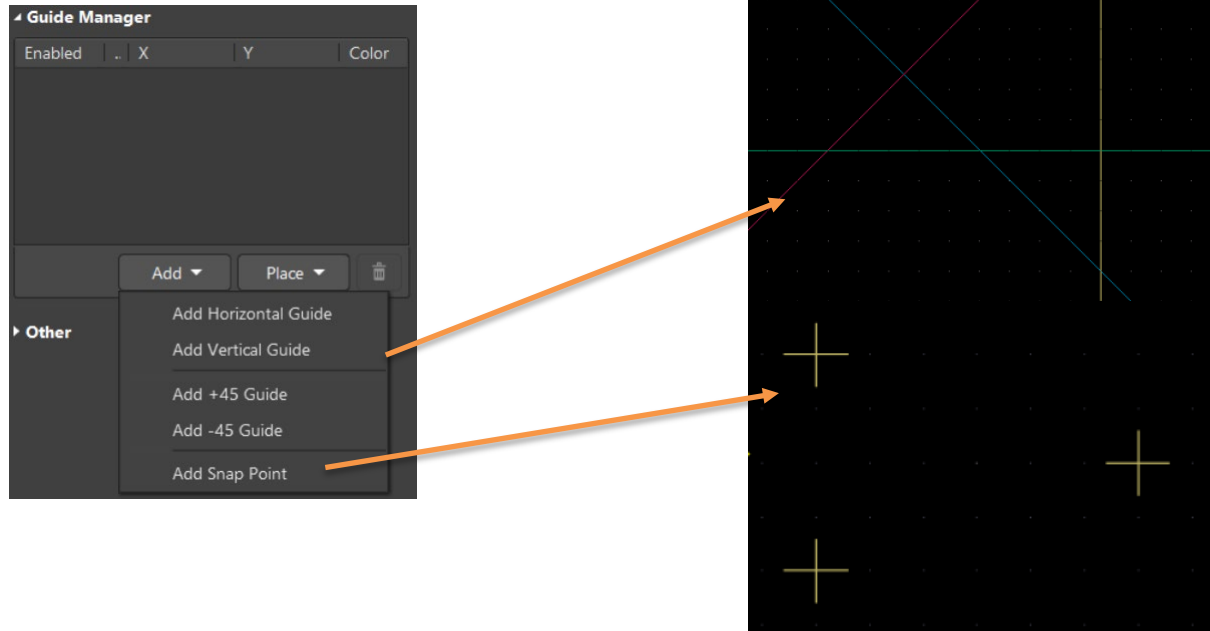
USER-DEFINABLE GRIDS

- Snap to Grid, could be linear or polar

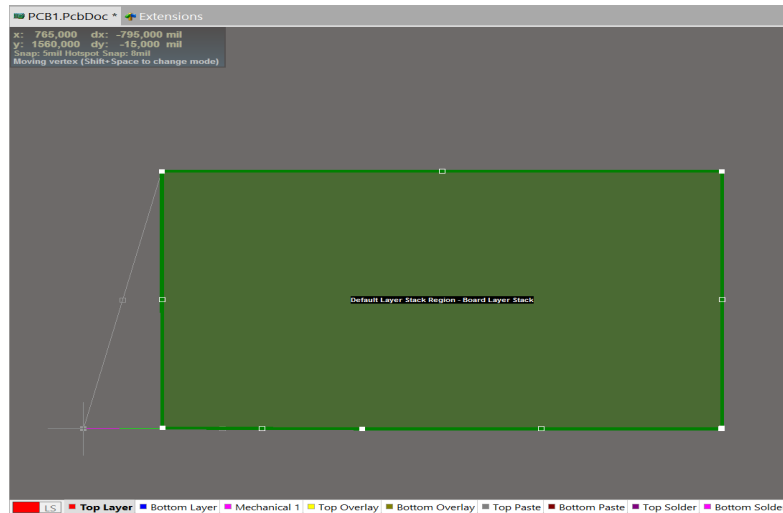
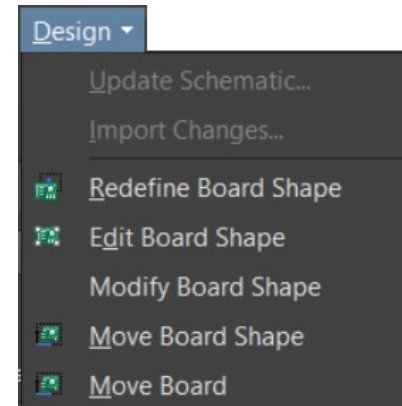
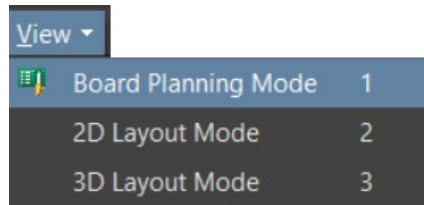


SNAP GUIDES

- Linear Snap or Point Snap

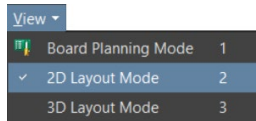


BOARD SHAPE OPERATIONS

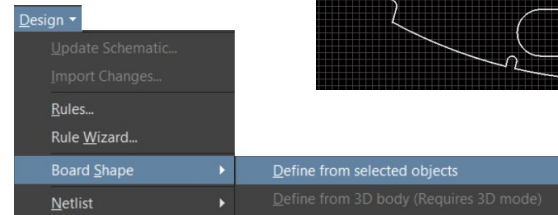
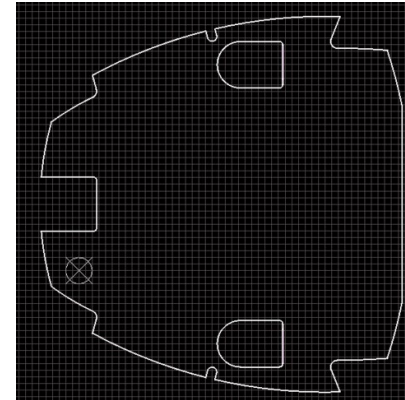
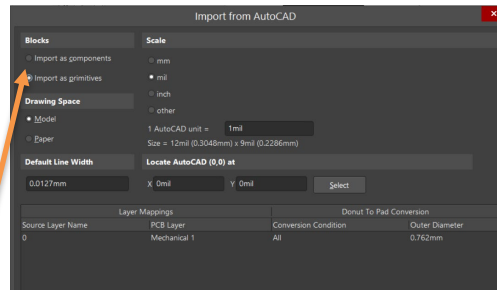
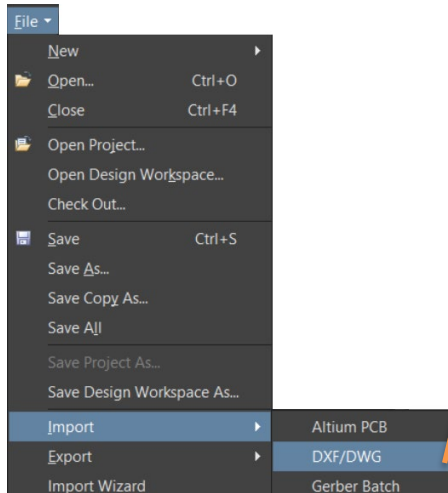


We can make some basic operations with the board shape area, from Design tool list, on Board Planning Mode view mode.

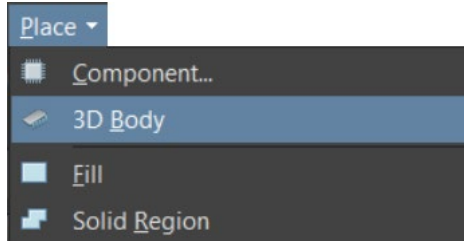
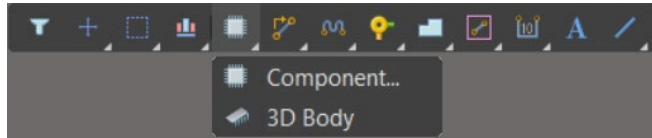
DEFINE BOARD SHAPE FROM 2D MODEL



Please note that the shape to be imported must be a closed shape, and internal cutouts are not automatically created (but can be defined later from imported objects).

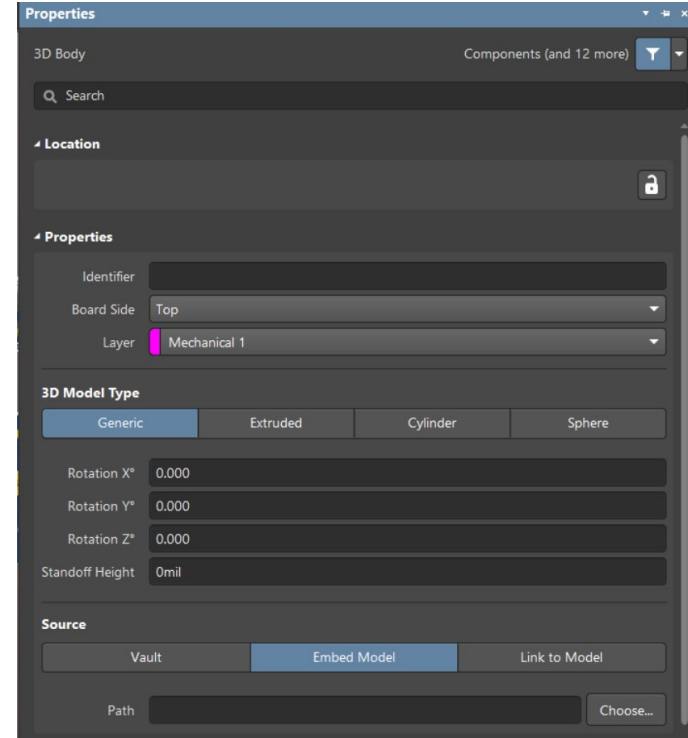


DEFINE BOARD SHAPE FROM 3D MODEL



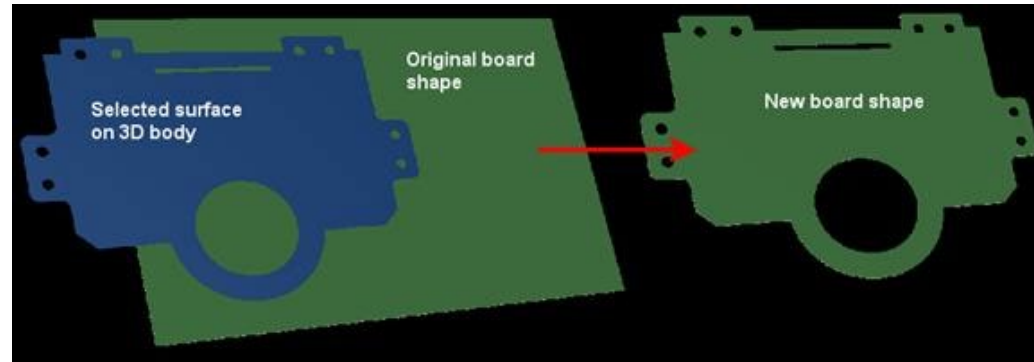
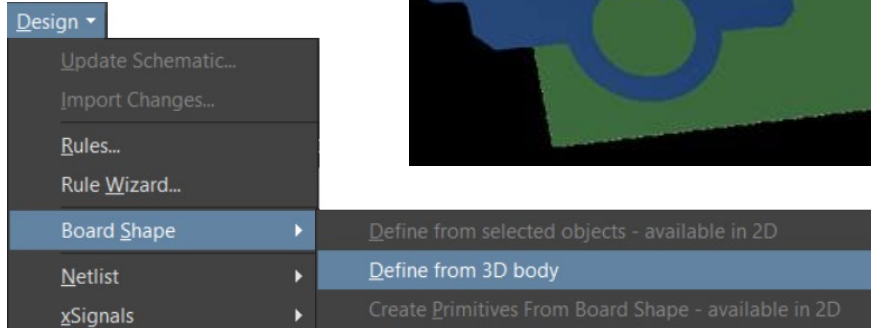
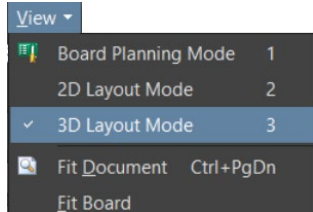
Before planning a new contour definition on our board shape from 3D body, we need to import a valid 3D model, from a previous design tool.

Altium works with step models.



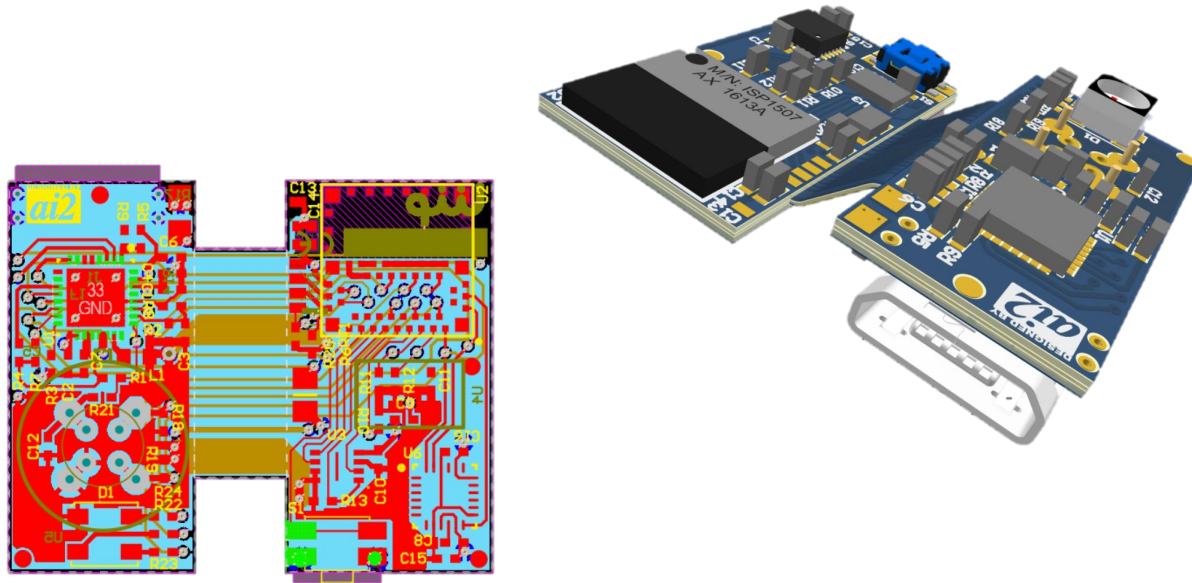
DEFINE BOARD SHAPE FROM 3D MODEL

Once 3D models have been imported to our PCB, we need to choose the valid model for our board shape.



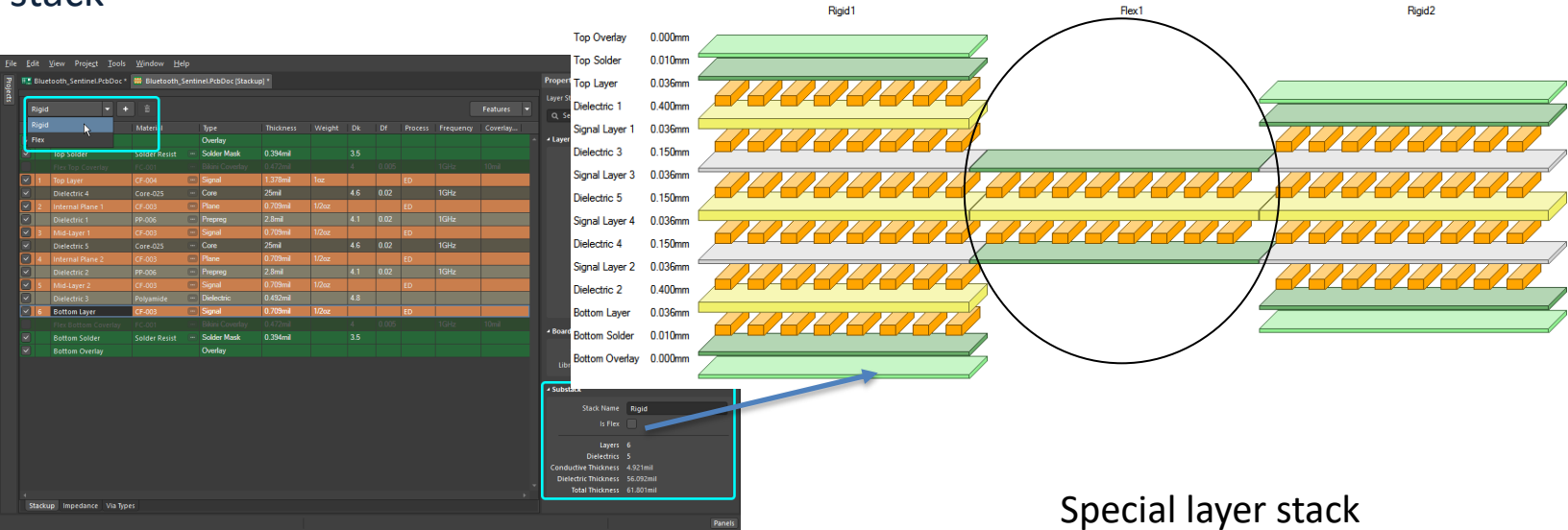
RIGI-FLEX BOARDS

To support the need to define a different set of layers in different areas of the board design, Altium supports the concept of multiple layer stacks.



RIGI-FLEX BOARDS

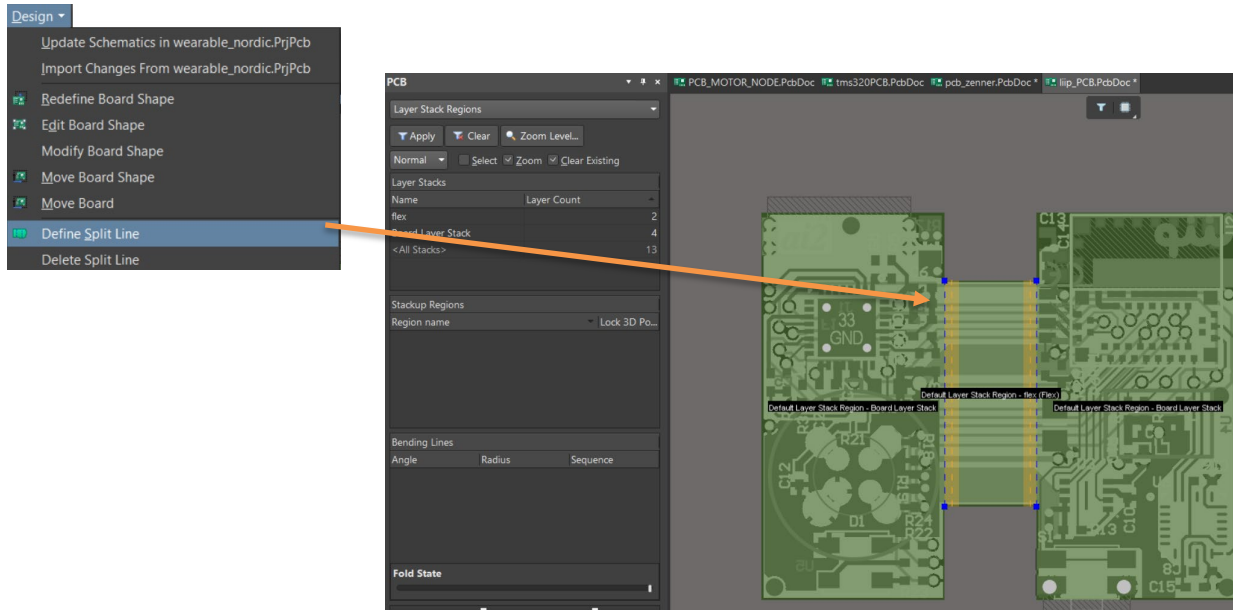
This is achieved by having an overall master layer stack that defines the total set of layers available to the board designer in this design. From this master layer stack, any number of sub-stacks can be defined, using any of the layers available in the master stack



Special layer stack for flex pcb zone.

RIGI-FLEX BOARDS

Defining a new board region or editing an existing region is done by switching the PCB editor to *Board Planning Mode* (**View » Board Planning Mode**, or press the **1** shortcut). You split the original, single board region into two regions by placing a *Split Line* (**Design » Define Split Line**)

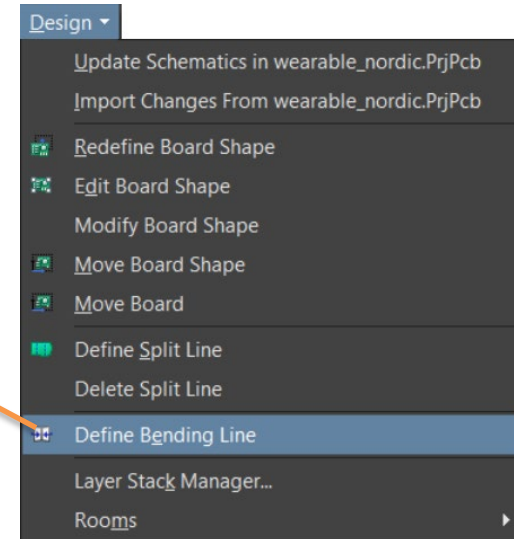
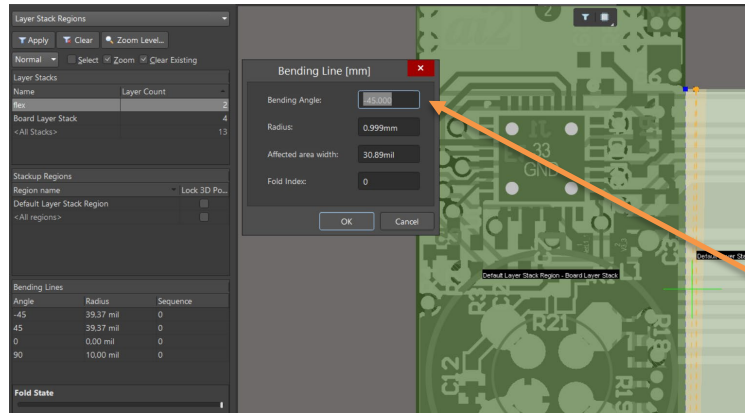


RIGI-FLEX BOARDS

Select the **Design » Define Bending Line** command and click inside the board region where the Bending Line is to be placed, one end of the line will attach to the edge of the region closest to the click location, the other end will attach to the cursor.

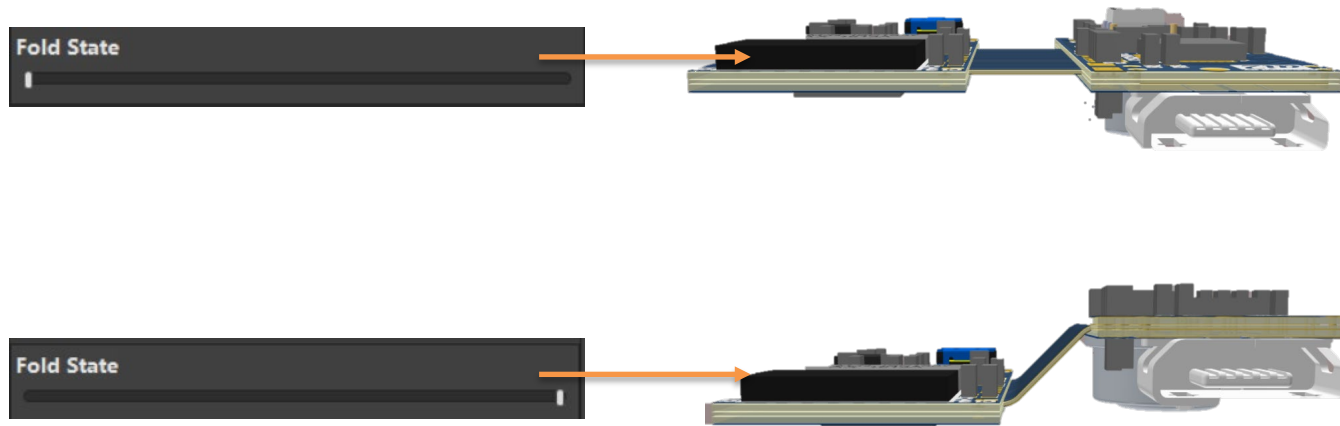
Then position the second end in the required location then click once to place it.

Double click on Bending Lines dialog (PCB Panel), and appear a new dialog box to configure folding angle, radius and index.



RIGI-FLEX BOARDS

We can revise the final design and bending angles, with a real visualization of the final design. Altium offer us a Fold State bar to visualize it, from PCB panel, when we select Layer Stack Regions.



DIMENSIONS

Place ▾

Component...

3D Body

Dimension

Work Guides

Linear

Angular

Radial

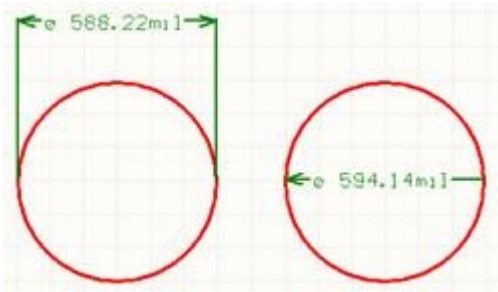
Ordinate

Leader

Standard



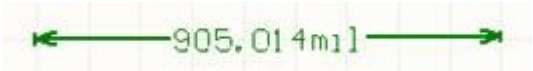
Leader



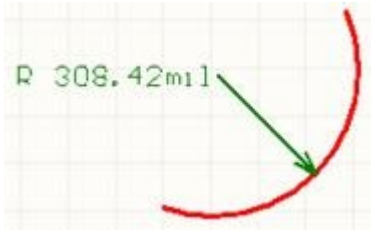
Linear Diameter



Linear



Standard



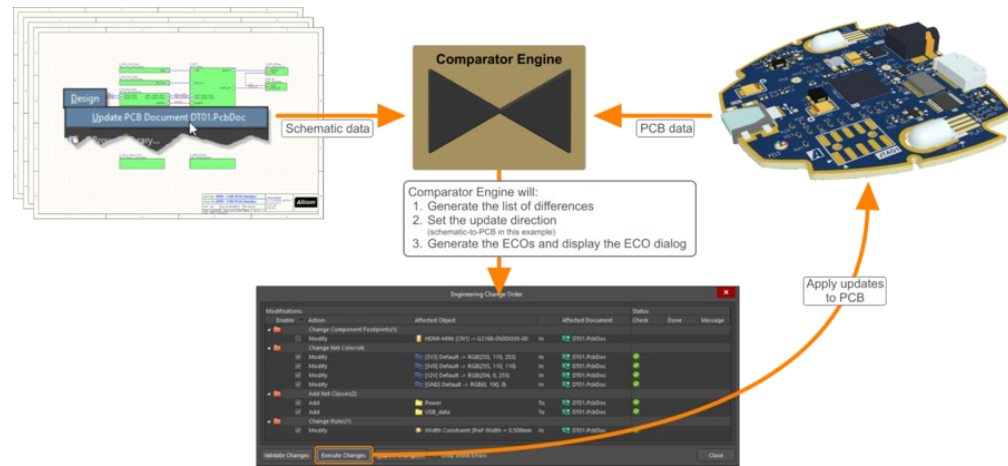
Radial



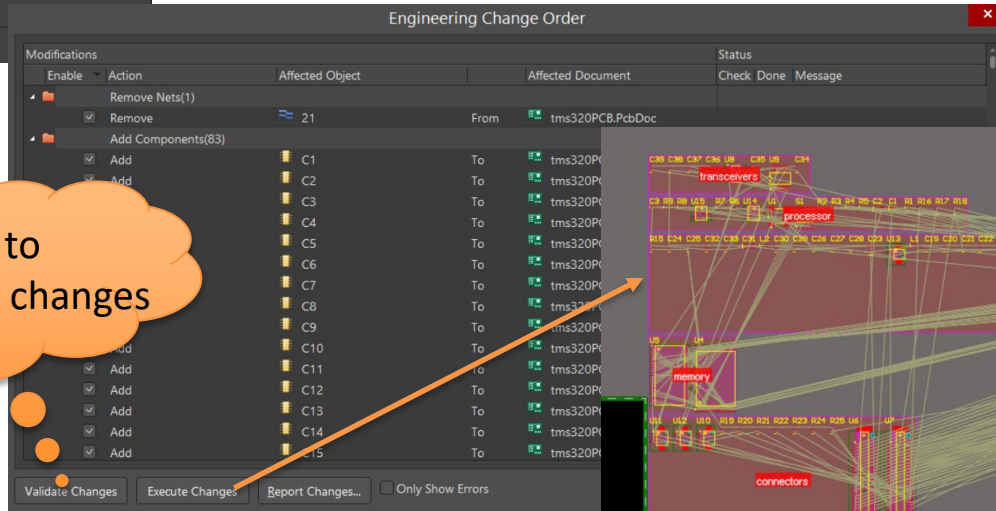
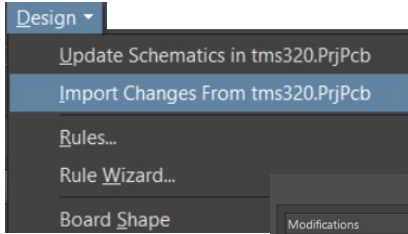
Angular

NETLIST IMPORT

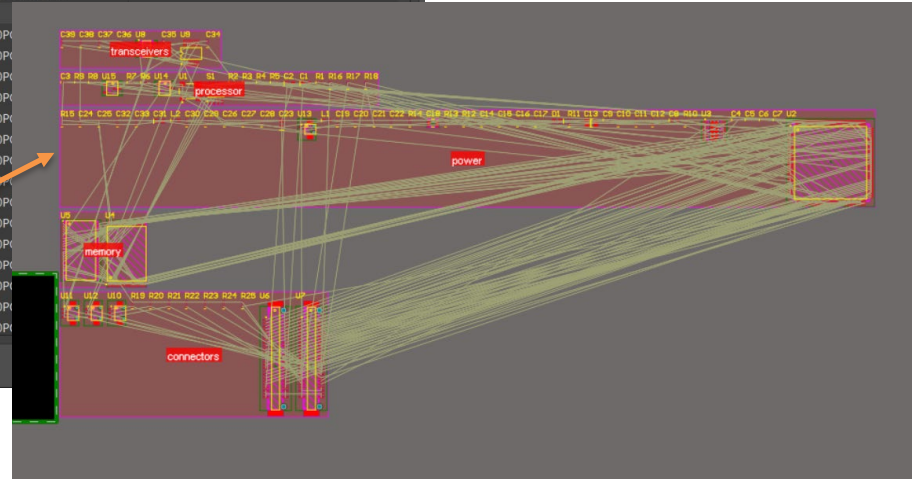
Whether you're transferring a captured design to a new PCB for the first time, or making changes to an existing design on either the schematic or PCB side, you need some way to keep the two sides in-sync. Altium Designer includes powerful design synchronization capabilities that simplify the task of keeping the design synchronized, allowing the designer to keep their focus on the creative aspects of the design process.



NETLIST IMPORT



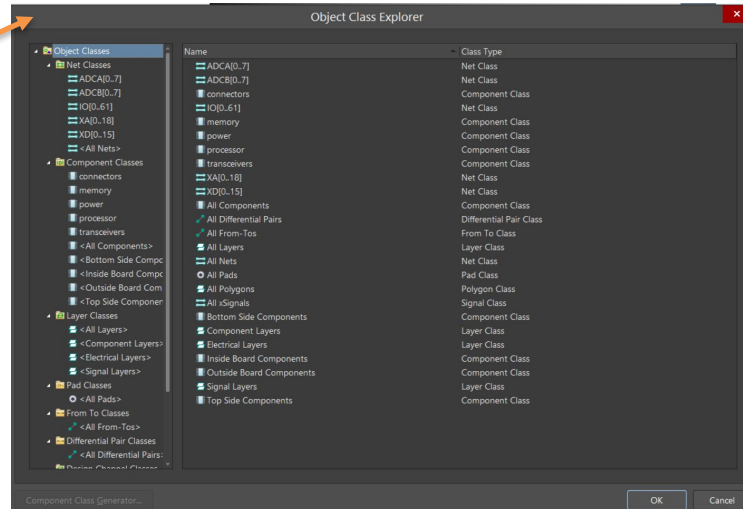
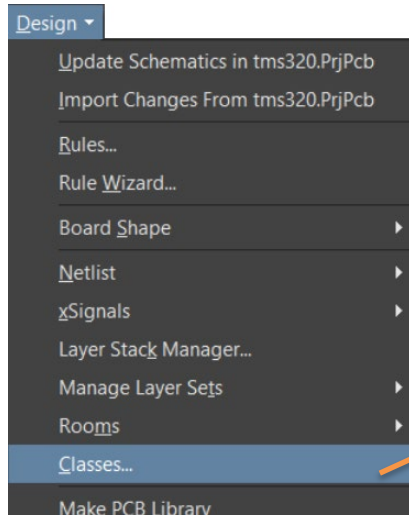
1º we need to validate the changes



WORKING WITH CLASSES

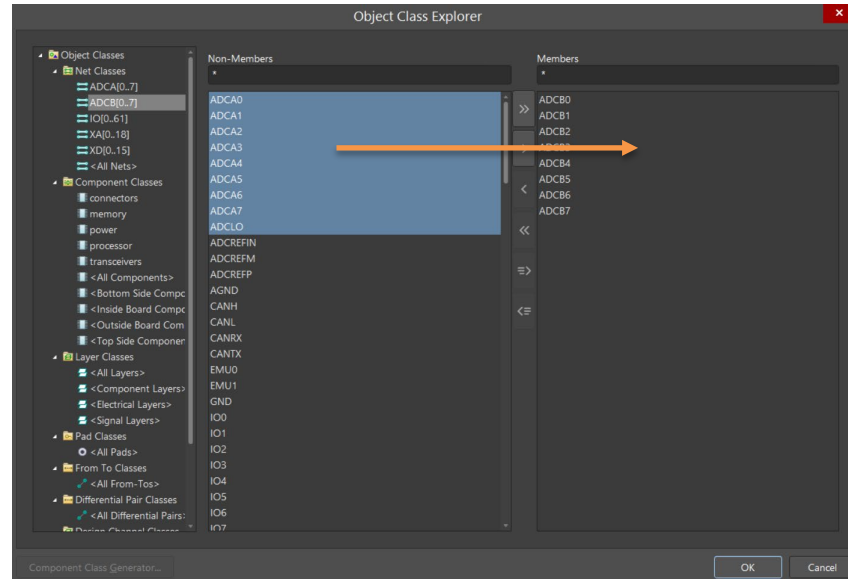
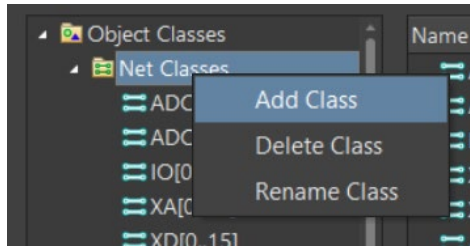
Altium Designer already provided high-quality, robust support for generation of classes when transferring the design from the Schematic to PCB, and then we can work directly from PCB area, from **Object Class Explorer**.

The classes in Altium give us the design support to work with groups of equal or different members, this ease us the PCB workflow.



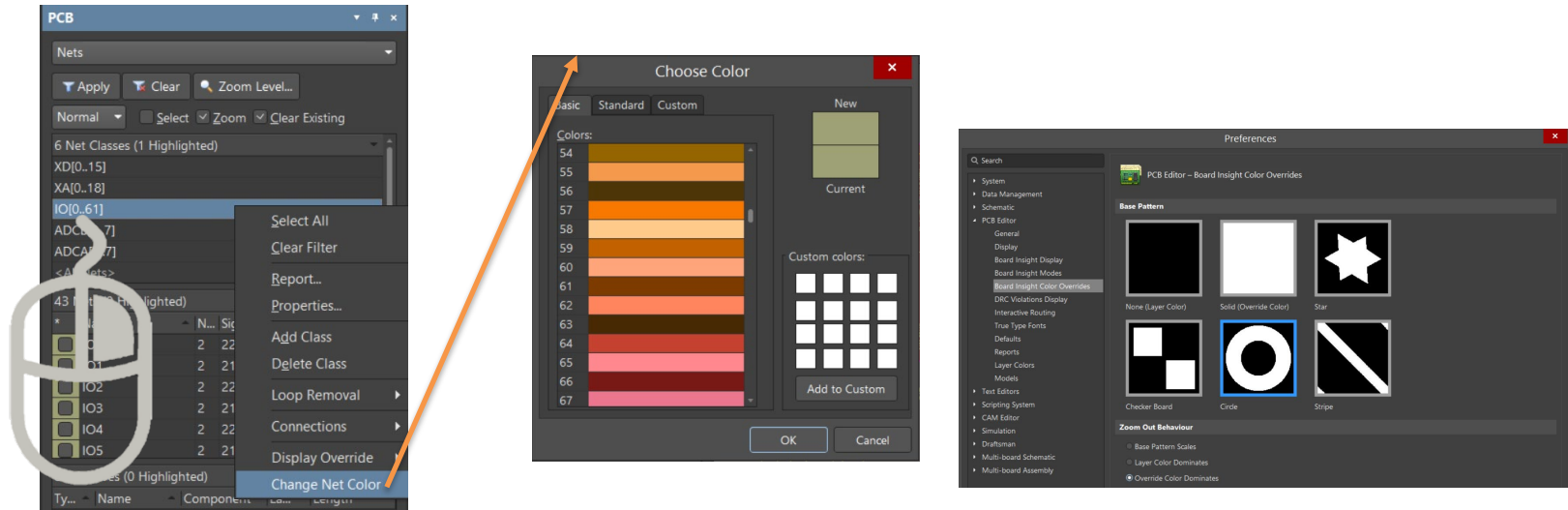
WORKING WITH CLASSES

We can integrate new clases on our PCB designs, adding new signals, components, poligons, etc to our different clases definition on the Object Class Explorer.



WORKING WITH CLASSES

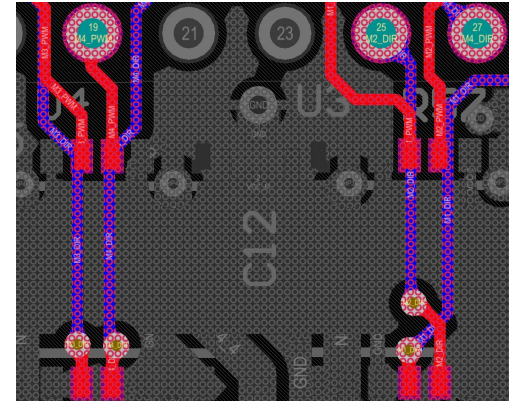
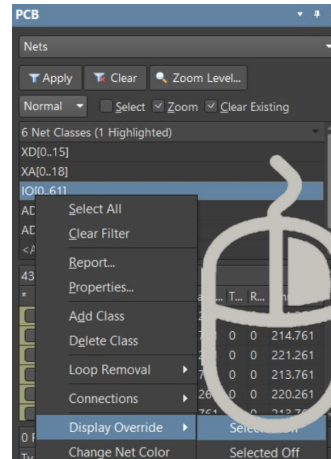
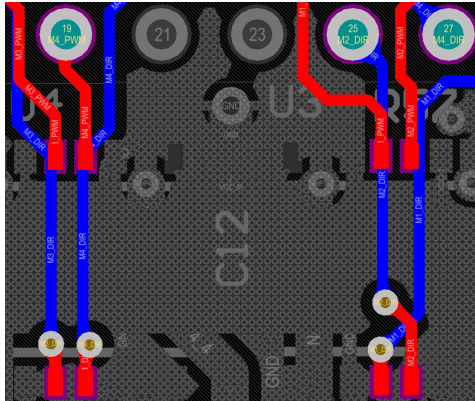
The use of classes give us a greater control over the highlighting of nets on your PCB documents, using your own **overriding color** scheme. Then, instead of purely having a net object colored using its respective layer color, you can assign to use a specific alternative color. Add to this a range of preconfigured color override patterns and you have a powerful enhancement to your PCB visualization arsenal.



WORKING WITH CLASSES

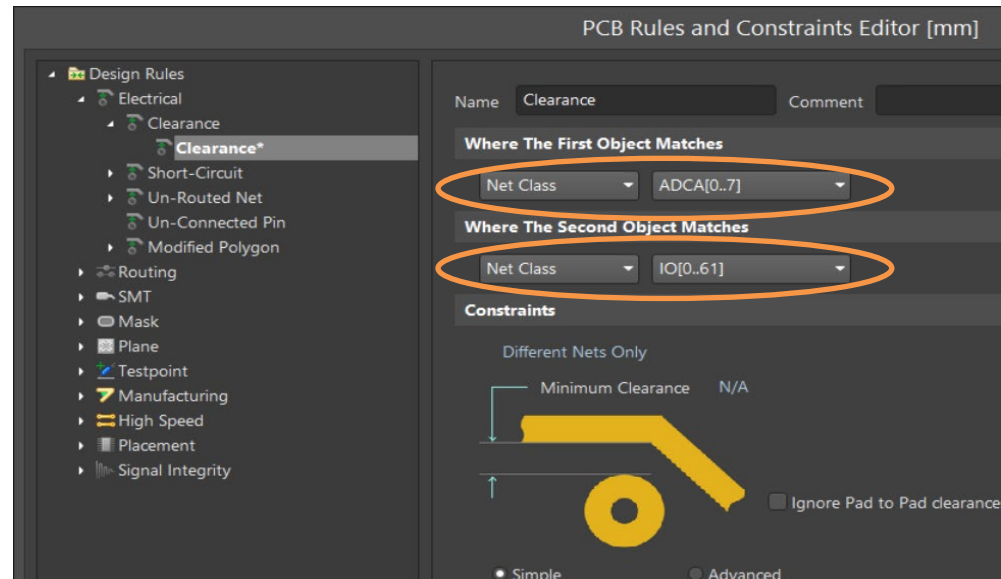
As with assigning colors to nets, you can enable/disable the net color override feature for an individual net, multiple selected nets, or all nets in one or more selected net classes. Simply make your selection accordingly and then Selected On-Off.

The enabled nets will be displayed in accordance with defined coloring and chosen override pattern.



WORKING WITH CLASSES

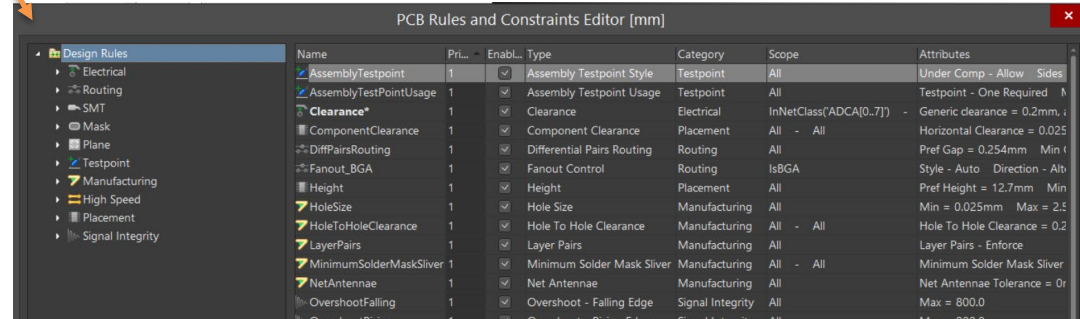
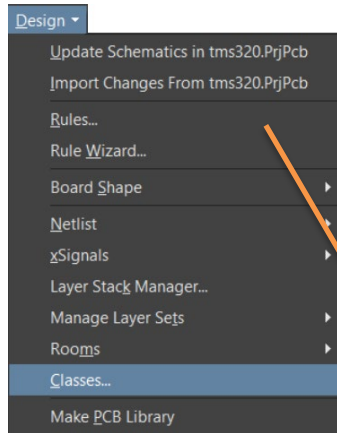
They can be used when scoping design rules to constrain the design in a specific way, or when filtering objects in the design workspace.



PCB RULES & CONSTRAINTS

The PCB board design could be a complex task, and we take in to account a great list of constraints in our designs:

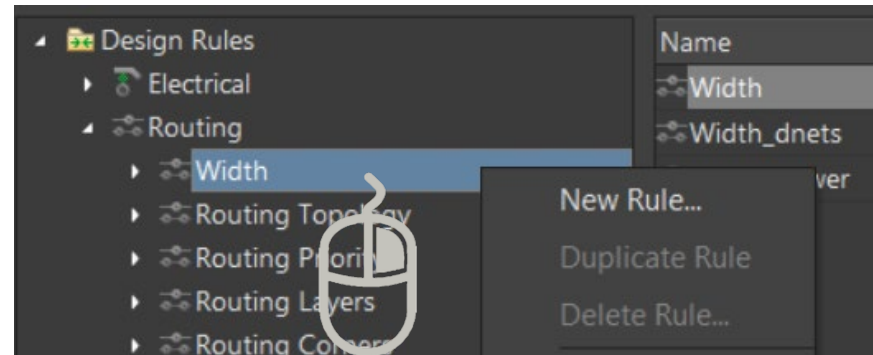
- PCB manufacturer.
- Design technology (impedances, distances)
- Test.
- Mechanical design interaction.



PCB RULES & CONSTRAINTS

Management of the design rules for the current board design, is performed from the *PCB Rules and Constraints Editor* dialog.

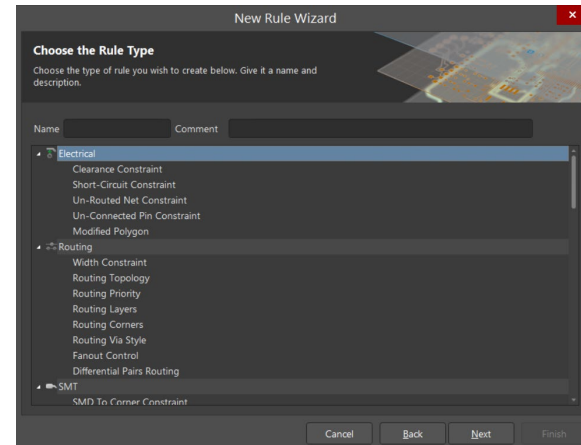
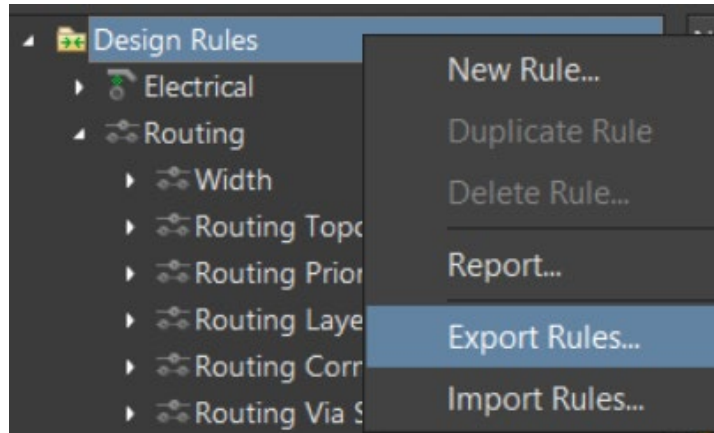
The folder-tree pane on the left side of the dialog, each of the ten design rule categories are listed under the **Design Rules** folder. Click on this root folder to access a summary of all specific rules that have been defined, for all design rule types, across all categories.



PCB RULES & CONSTRAINTS

Design rules can be exported from, and imported to, the *PCB Rules and Constraints Editor* dialog. This allows you to save and load favorite rule definitions between different designs.

For a first steps on a design you can use a Rule Wizard



PCB RULES & CONSTRAINTS

Electrical rules:

This rules applies to **different nets, tracks and polygons**, on cooper layers.

SMT rules:

This rules applies to **different smt pads**, on cooper layers.

Plane rules:

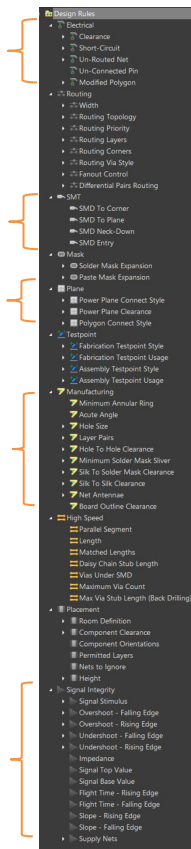
This rules applies between **different polygons and TH pads or vias**, on cooper layers.

Manufacturing rules:

This rules come from the PCB manufacturer company.

Signal Integrity rules:

This rules are used by SI simulator to compare as a max error values.



Routing rules:

This rules applies between **different nets, tracks and vías**, on cooper layers.

Mask rules:

This rules applies to **different pads**, on cooper layers.

Testpoint rules:

This rules applies to **different pads**, on cooper layers. Define the distances to use on defined testpoints.

High Speed rules:

This rules applies to **different Nets**, controlling lengths, matching, etc.

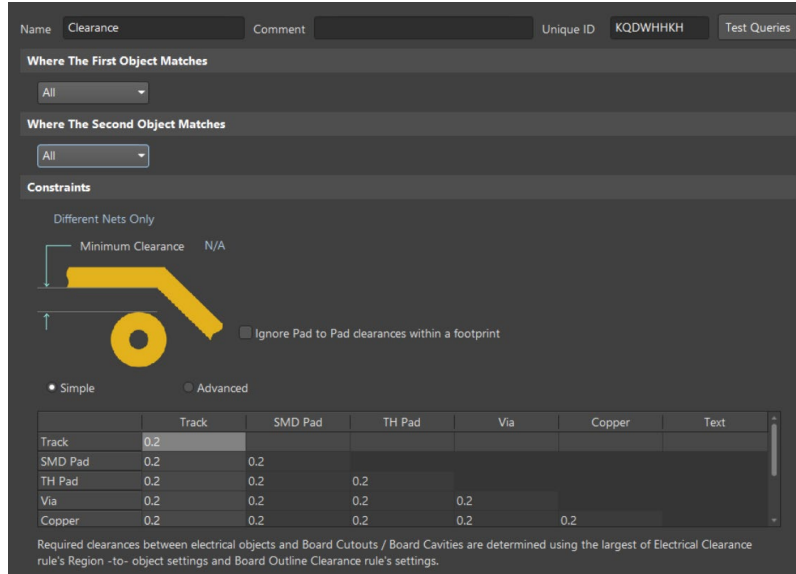
Placement rules:

This rules applies to **different components**, on different layers.

PCB RULES & CONSTRAINTS

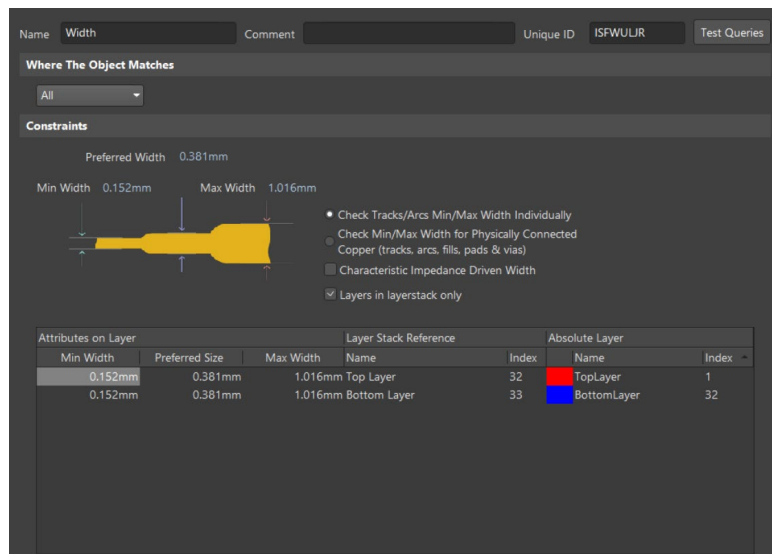
From Clearance rule, we can define the minimum distance between different objects on our board. Usually this values come from our pcb supplier.

Use the matrix to fine tune clearances between any objects in the design. In combination with rule-scoping, you have all the flexibility you need to build a concise and targeted set of rules to meet even the most stringent of clearance needs.



PCB RULES & CONSTRAINTS

Width rule defines the width of tracks placed on the copper (signal) layers.

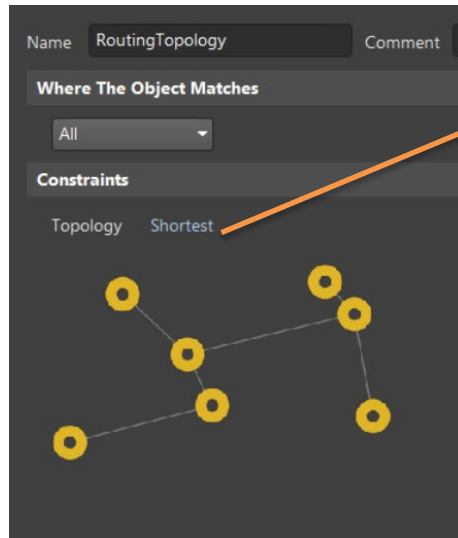


- **Check Tracks/Arcs Min/Max Width Individually** – checks individual widths of tracks and arcs fall within the minimum and maximum range.

- **Check Min/Max Width for Physically Connected Copper (tracks, arcs, fills, pads & vias)** – checks the width of routed copper formed by a combination of tracks, arcs, fills, pads & vias falls within the minimum and maximum range.

PCB RULES & CONSTRAINTS

This rule specifies the topology to be employed when routing nets on the board. The topology of a net is the arrangement or pattern of the pin-to-pin connections.

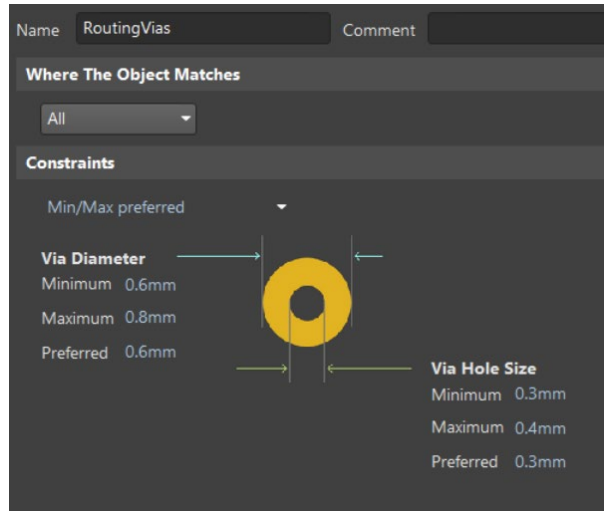


- **Shortest:** the shortest overall connection length
- **Horizontal:** force routing in the horizontal direction
- **Vertical:** force routing in the vertical direction.
- **Daisy-Simple:** chains all the nodes together, one after the other.
- **Daisy-MidDriven:** places the source node(s) in the center of the daisy chain, divides the loads equally and chains them off either side of the source(s).
- **Daisy-Balanced:** connect to the source in a star pattern.
- **Starburst:** connects each node directly to the source node.

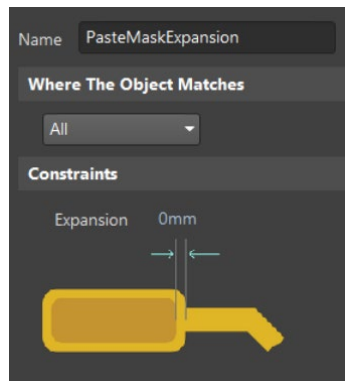
PCB RULES & CONSTRAINTS

The **Minimum** and **Maximum** via attributes are obeyed by the Online DRC and Batch DRC. They also determine the range of permissible values that can be used during interactive routing.

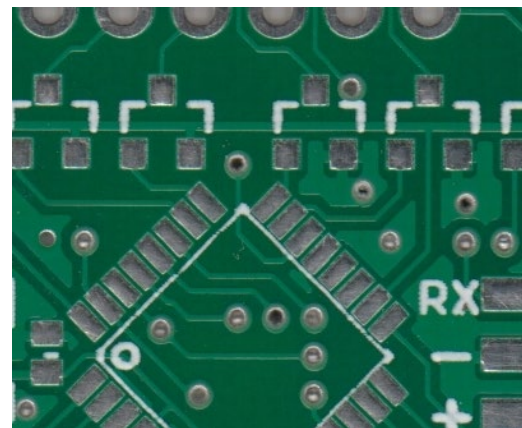
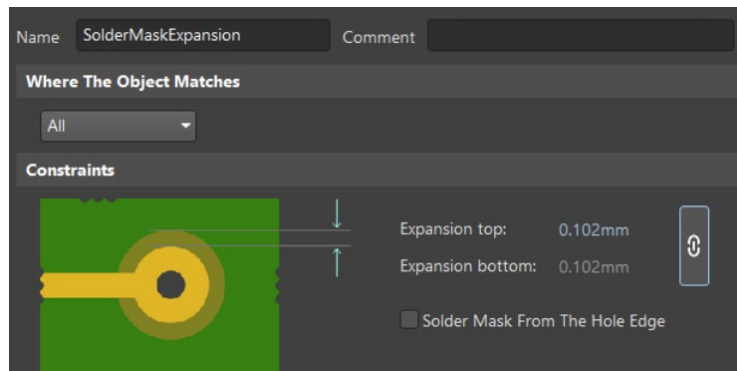
The **Preferred** via attributes are used by the Autorouter



PCB RULES & CONSTRAINTS

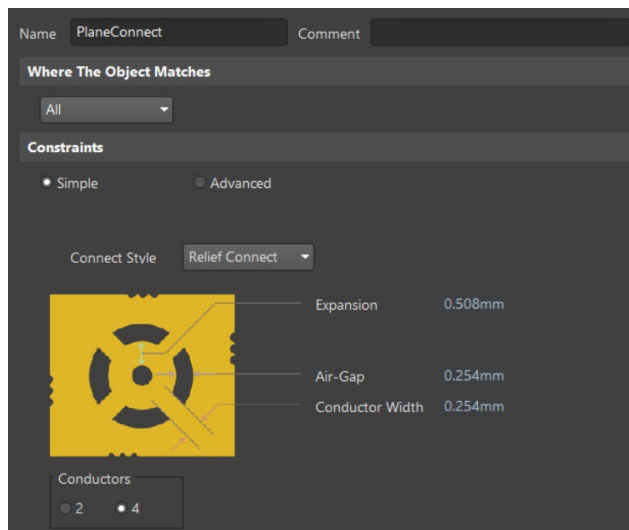


The paste and solder mask expansion can be defined for pads on an individual basis, in the associated properties dialog. Options are available to follow the expansion defined in the applicable design rule, or to override the rule and apply a specified expansion directly to the individual pad in question.



PCB RULES & CONSTRAINTS

Defines the style of the connection from a pin of a component, targeted by the scope of the rule, to a polygon plane. The following three styles are available:

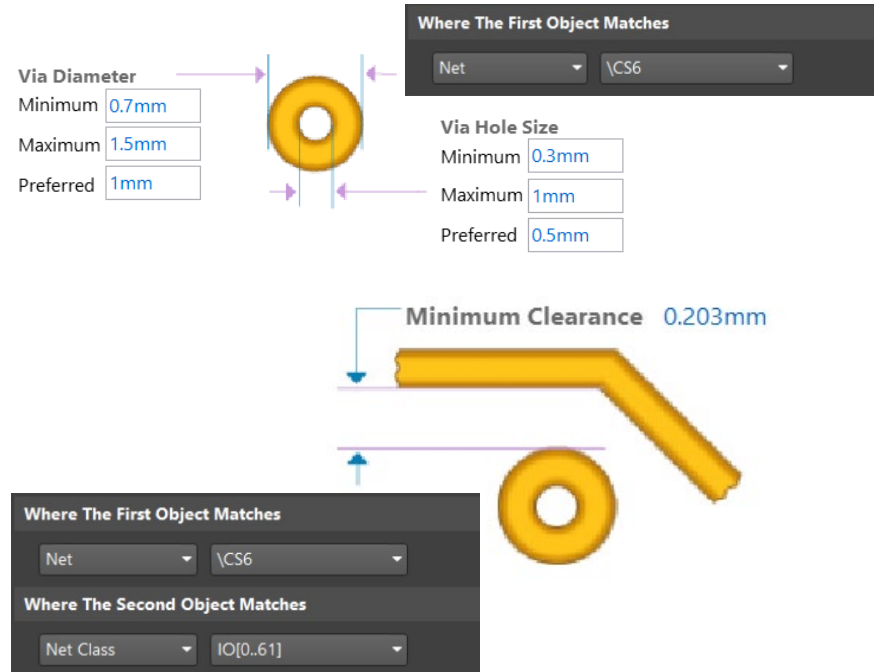


- **Relief Connect:** connect using a thermal relief connection.
- **Direct Connect:** connect using solid copper to the pin.
- **No Connect:** do not connect a component pin to the polygon plane.

PCB RULES & CONSTRAINTS

There are two types of design rules

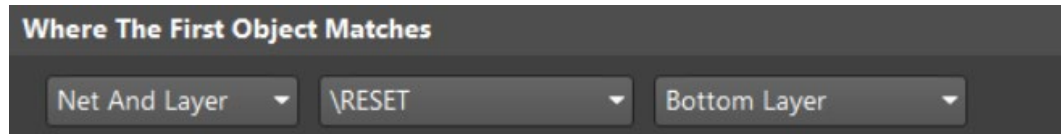
- **Unary rules** apply to one object, or each object in a set of objects. As a consequence, unary design rules have one rule scope.
- **Binary rules** apply between two objects, or between any object in one set to any object in a second set. As a consequence, binary design rules have two rule scopes.



PCB RULES & CONSTRAINTS

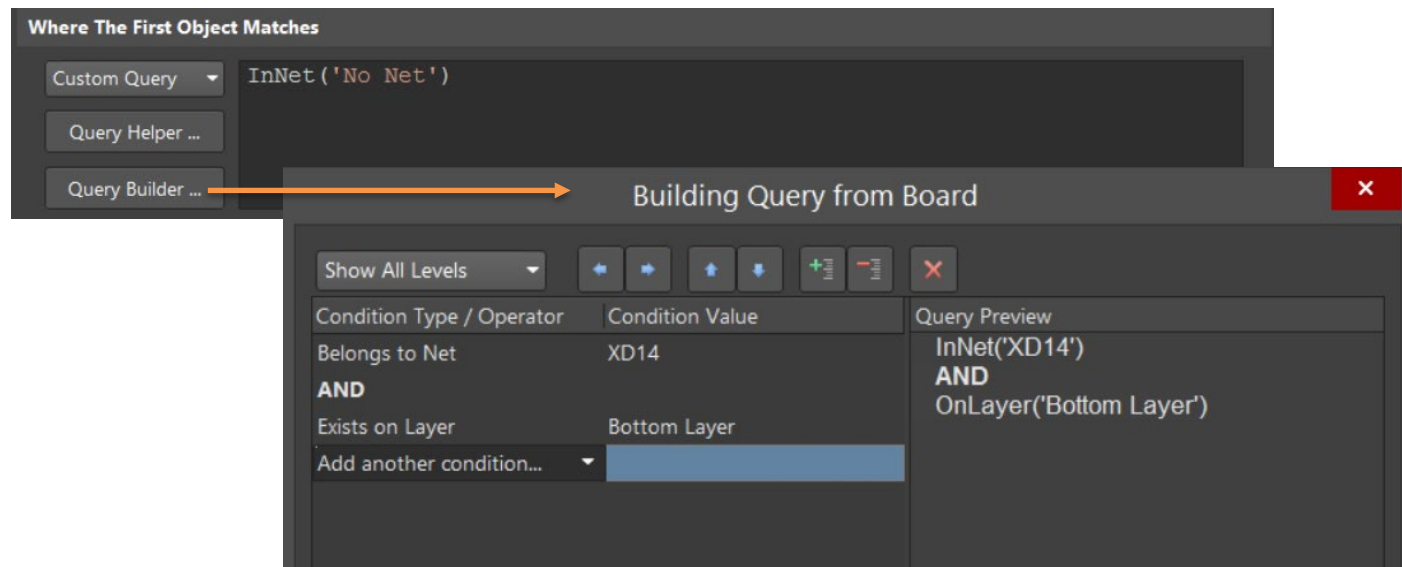
When defining the scope of a design rule you are essentially building a query to define the member objects that are governed by the rule. Use the options available in the dialog to build the query required. Depending on whether the rule is unary or binary, you will need to define one or two scopes respectively.

- all design objects.
- all objects in a specific net.
- all objects in a specific net class.
- all objects on a specific layer.
- all objects in a specific net and on a specific layer.
- all defined nets in the design.
- all defined net classes in the design or.
- all currently enabled layers in the design.
- etc



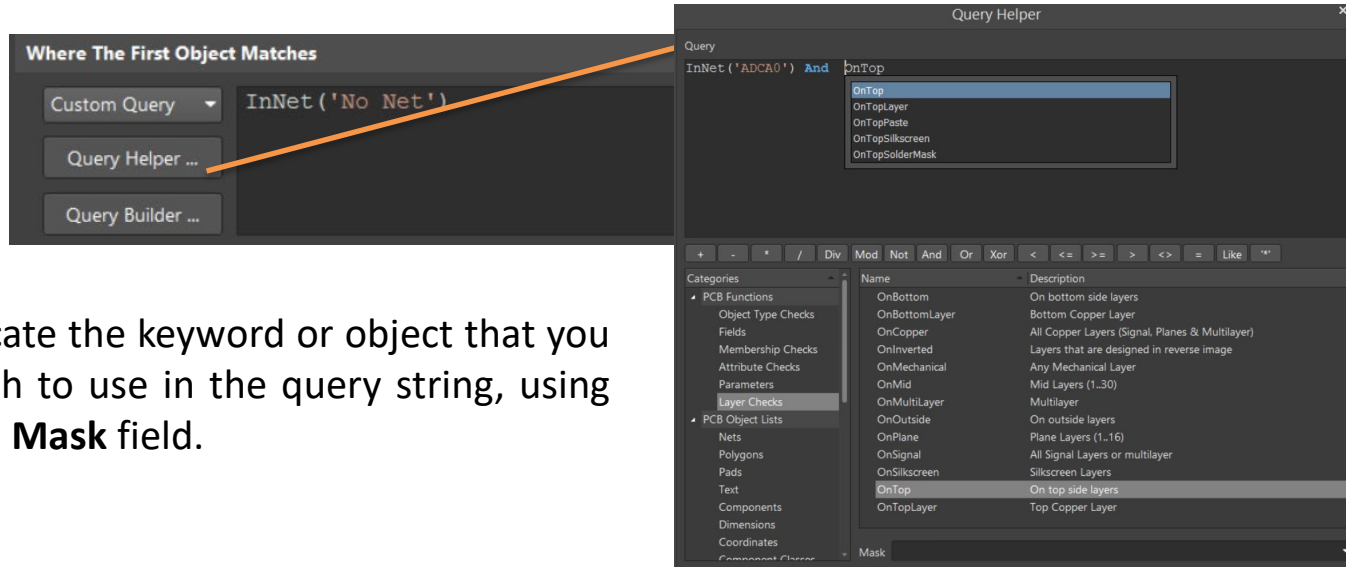
PCB RULES & CONSTRAINTS

Building Query from Board dialog enables you to create a query for targeting specific objects in the design document, by simple construction of a string of ANDed and/or ORed conditions. The left-hand section of the dialog is where you specify the condition(s) that you require to target the set of objects needed.



PCB RULES & CONSTRAINTS

The underlying query engine analyzes the PCB design and lists all available objects, along with generic keywords for use in queries. Use the **Query** region of the dialog to compose a query expression. You can type directly within the region. As you type, a context-sensitive prompt list of possible keywords or objects will appear as an aid.

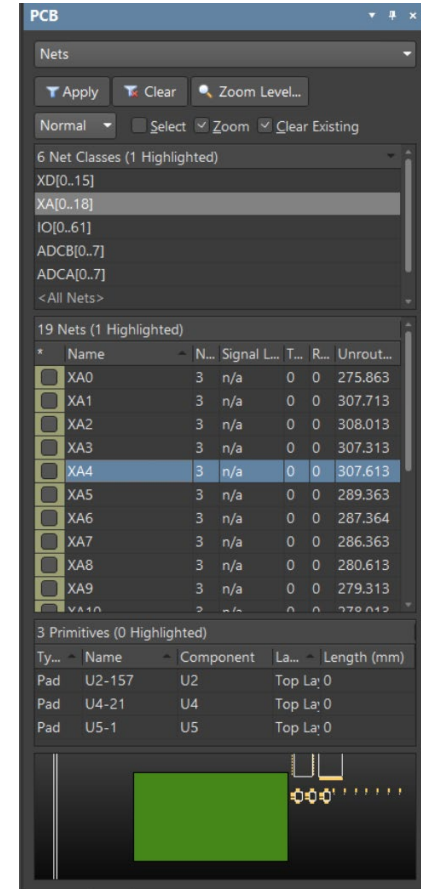


Locate the keyword or object that you wish to use in the query string, using the **Mask** field.

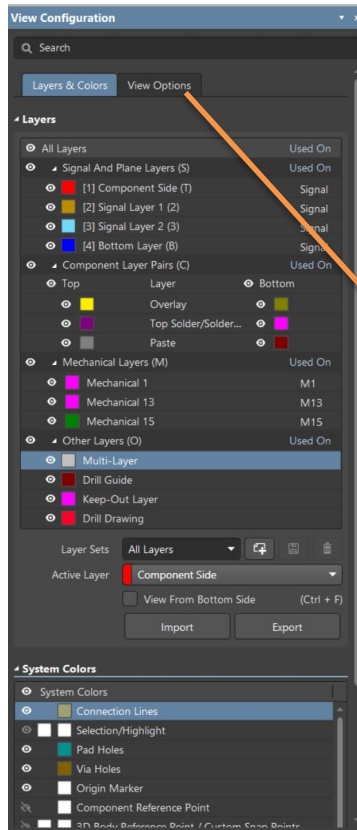
PCB PANEL

The PCB panel allows you to browse the current PCB design using various filter modes to determine which object types or design elements are listed, highlighted or selected.

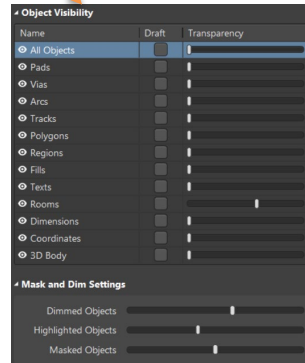
- The top drop-down list in the panel is where you select the panel mode. You can define a browsing mode - either nets, components, design rules, 3D models (3D bodies) or polygons - or you can use the From-To, Split-Plane, Differential Pairs, Pads&Vias, and Hole Size Editors.
- The bottom window shows the whole board and the current position of the main design window



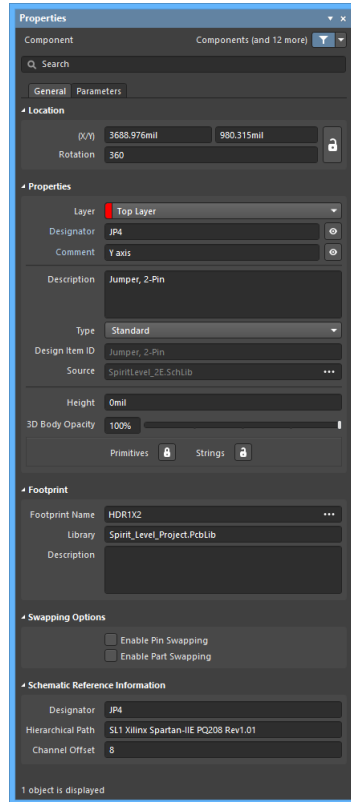
VIEW CONFIGURATION PANEL



The *View Configuration* panel is used to configure what is currently displayed in the workspace and how it is displayed. This includes layer visibility and color, object visibility and transparency, masking and dimming levels, the current single layer mode, and a number of additional workspace display features, such as the display of net names on pads, via and tracks.



PROPERTIES PANEL



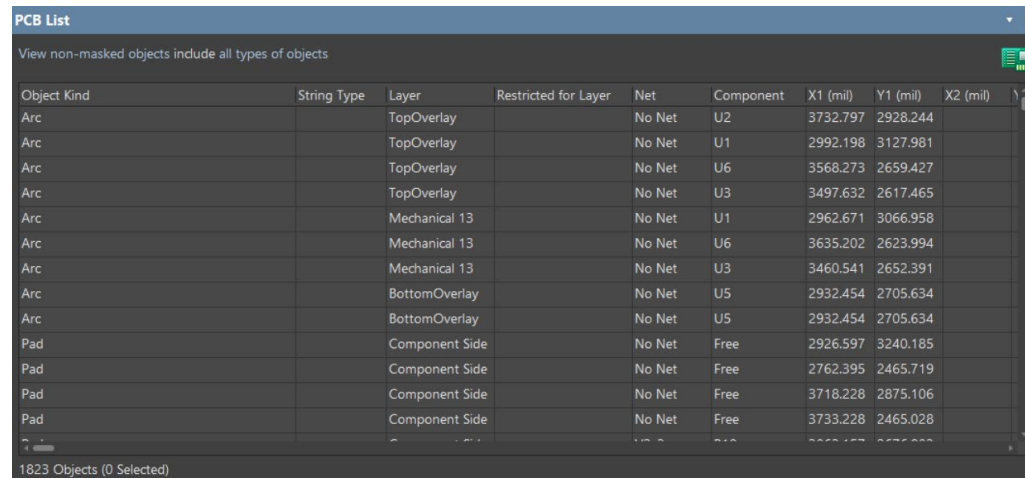
When a design object is selected, the panel will present options specific to that object type. The following table lists the object types available for placement within a PCB document - click a link to access the properties page for that object.

PCB Editor object properties are definable options that specify the visual style, content and behaviour of the placed object.

PCB LIST PANEL

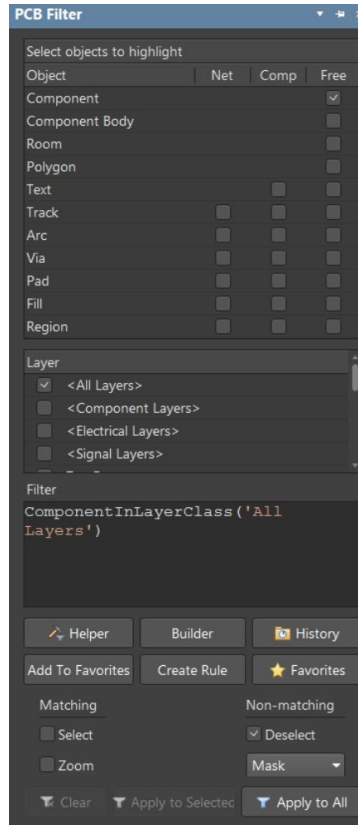
The PCB List panel allows you to display design objects from the active document in tabular format, enabling you to quickly inspect and modify object attributes.

Is a useful tool to change a lot of properties in a large number of objects in an easy way.



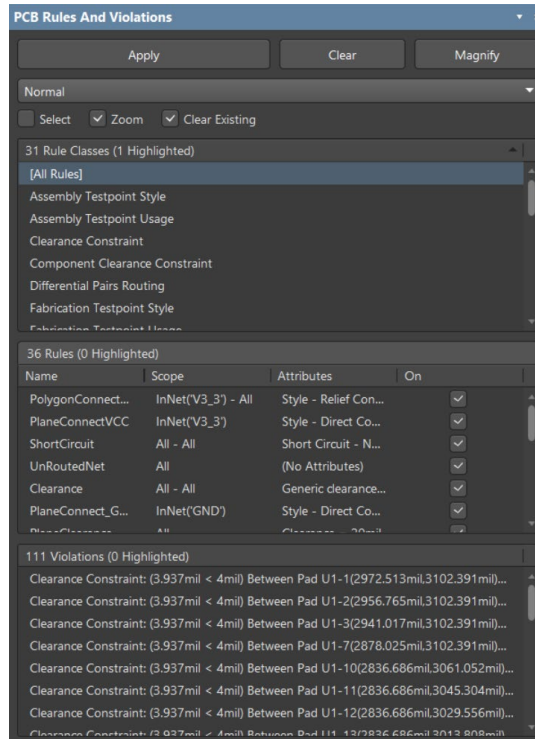
View non-masked objects include all types of objects								
Object Kind	String Type	Layer	Restricted for Layer	Net	Component	X1 (mil)	Y1 (mil)	X2 (mil)
Arc		TopOverlay		No Net	U2	3732.797	2928.244	
Arc		TopOverlay		No Net	U1	2992.198	3127.981	
Arc		TopOverlay		No Net	U6	3568.273	2659.427	
Arc		TopOverlay		No Net	U3	3497.632	2617.465	
Arc		Mechanical 13		No Net	U1	2962.671	3066.958	
Arc		Mechanical 13		No Net	U6	3635.202	2623.994	
Arc		Mechanical 13		No Net	U3	3460.541	2652.391	
Arc		BottomOverlay		No Net	U5	2932.454	2705.634	
Arc		BottomOverlay		No Net	U5	2932.454	2705.634	
Pad		Component Side		No Net	Free	2926.597	3240.185	
Pad		Component Side		No Net	Free	2762.395	2465.719	
Pad		Component Side		No Net	Free	3718.228	2875.106	
Pad		Component Side		No Net	Free	3733.228	2465.028	
1823 Objects (0 Selected)								

PCB FILTER PANEL



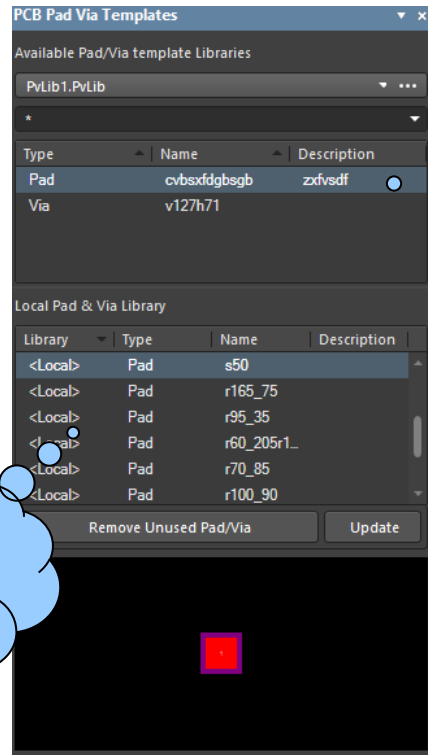
The PCB Filter panel allows you to construct filters through the creation of logical queries. A defined filter can then be applied to the active PCB document, allowing you to select and edit multiple objects with great accuracy and efficiency.

PCB RULES AND VIOLATION PANEL



Design Rule Checking (DRC) is a powerful automated feature that checks both the logical and physical integrity of your design. The *PCB Rules And Violations* panel allows easy browsing of the enabled design rules and violations in the current board layout workspace. The panel provides a central point for browsing and editing rules, instigating design rules checks, and viewing individual violations as graphic overlays in the editor workspace.

PCB PAD VIA TEMPLATE PANEL



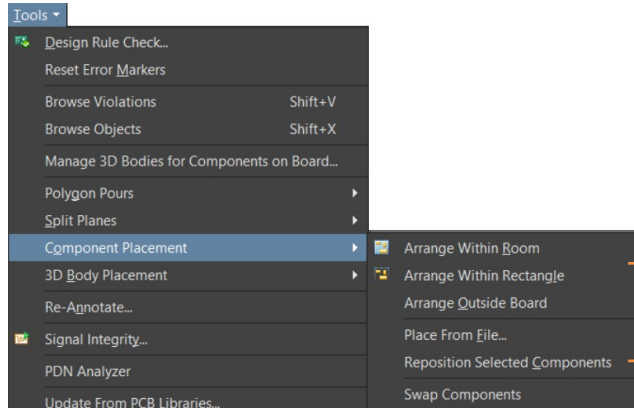
Pads on
library
selected

Pads on our
design

The *PCB Pad Via Templates* panel is a specialized panel that lists the pad/via templates that are assigned to the current PCB document (Local) or those available from Pad Via Libraries that have been installed or included within the current design project (Available libraries).

COMPONENT PLACEMENT

As all automatic or semi-automatic tools on Altium, their requires a lot of time for running properly, but we can find smart tools that help us to increase our design productivity with component placement task.



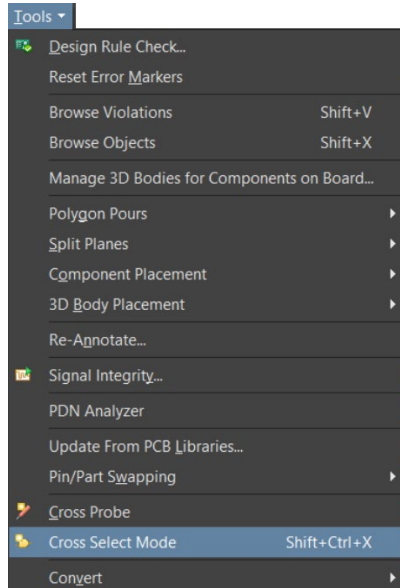
TIP: When select a component, with L we change their layer placement.

Automatic or semi-automatic placement tools (rooms based, from files, etc).

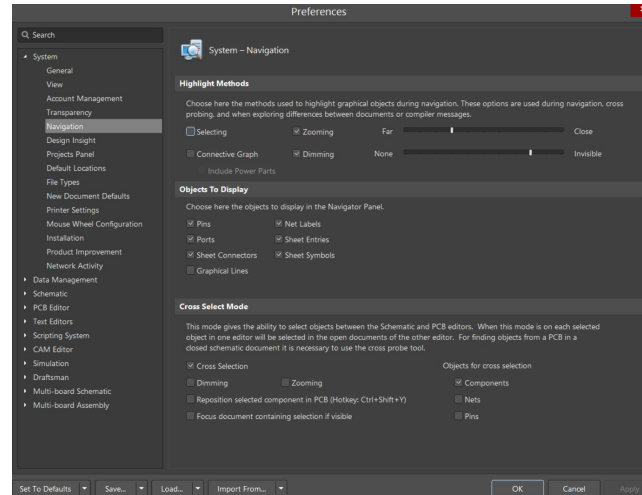
Manual placement tool.

COMPONENT PLACEMENT

If we need increase our placement productivity, we can directly select the components by design groups on schematic, and then on PCB board we can place them automatically.

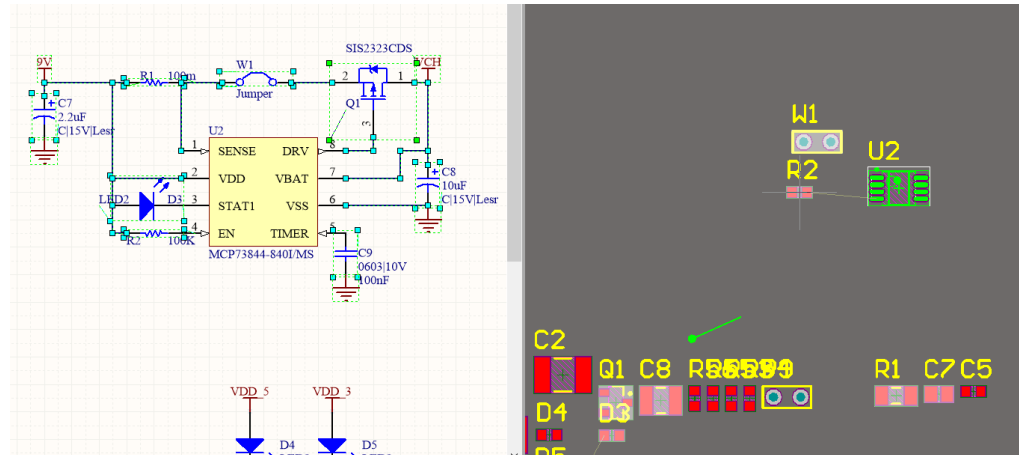
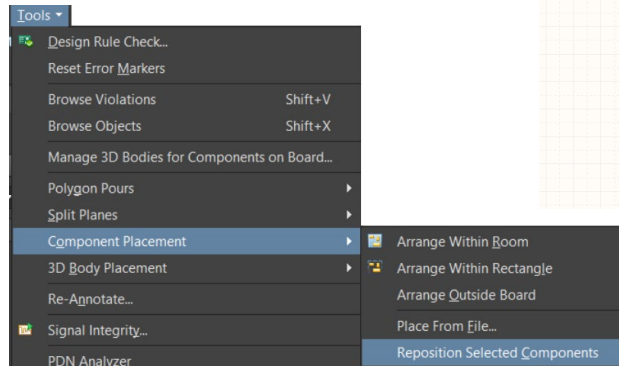


The Continuous Cross-Probing Mode allows you to stay in the source document while cross-probing to different objects on the target document. For this mode, ensure that the schematic and PCB documents are open side-by-side in the main design window.



COMPONENT PLACEMENT

Reposition Selected Components command is used to reposition a number of selected components sequentially, in the order that you selected them. You need to select the components from PCB or SCH.



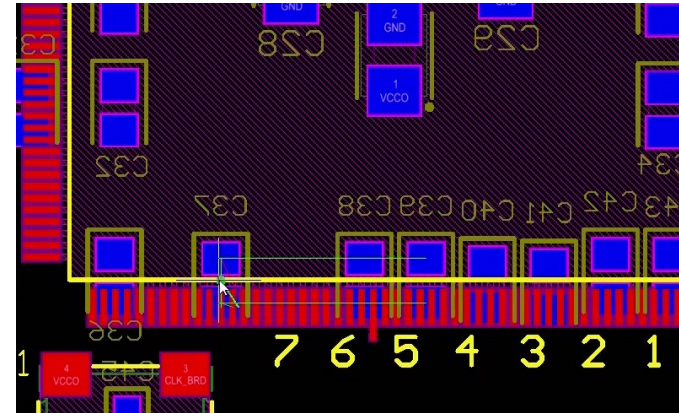
COMPONENT PLACEMENT

Component placement allows for dynamic alignment of components when dragging. Green indicator lines appear when the component is aligned either with adjacent components or with the pads of adjacent components.

Press and hold the **Ctrl** key during component placement to view alignment lines in relation to the boundaries of nearby components.

Press and hold the **Shift** key during component placement to view alignment lines by pads of nearby components.

You can cycle through placement modes (**Ignore, Push or Avoid Obstacles**) by pressing the **R** key during placement.



Then you can select a pair of components and easily swap their positions. After selecting the components you wish to swap, right-click and select **Component Actions » Swap Components**.

COMPONENT PLACEMENT

The **Copy Room Format** tool (Design\Rooms\Copy Room Formats) allow us to copy directly from a finish room the complet format placement, routing and properties from the source room selected.

YEL3_TR5 YEL3_LED5 YEL3_R5
YEL2_TR5 YEL2_LED5 YEL2_R5
YEL1_TR5 YEL1_LED5 YEL1_R5
RED4_TR1 RED4_LED1 RED4_R1
RED3_TR1 RED3_LED1 RED3_R1
RED2_TR1 RED2_LED1 RED2_R1
RED1_TR1 RED1_LED1 RED1_R1

ROOM1 ROOM2

Confirm Channel Format Copy

Source Room: ROOM1
Destination Room: ROOM2

Options

- ☒ Copy Component Placement
- ☒ Copy Designator & Comment Formatting
- ☒ Copy Routed Nets
- ☒ Copy Room Size/Shape
- ☐ Copy Selected Objects Only

Touching Objects Options

These options will apply to all primitives touching the source room

- ☐ Copy All Objects Touching the Room
- ☒ Exclude Nonnet Objects

Fully Endused Only / Endused & Touching

- ☒ Fully Endused Objects Only
- ☐ Endused & Touching Objects

Channel to Channel Component Matching

Match Components By Channel Offsets

Remove affected connections

Contained parts of connections only

Apply To Specified Channels

Information

Updated 0 Component(s) in 1 room(s)

OK

RED1_TR1 RED1_LED1 RED1_R1

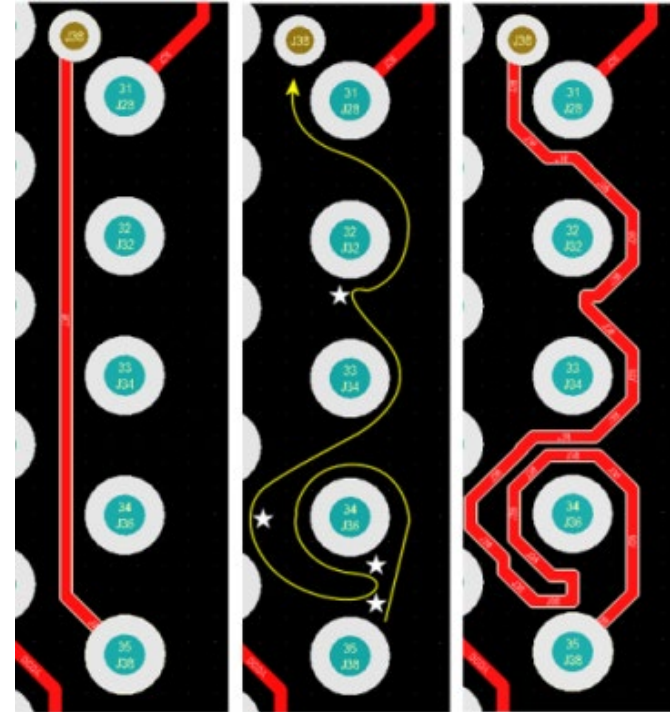
YEL2_TR5 YEL2_LED5 YEL2_R5
YEL1_TR5 YEL1_LED5 YEL1_R5
RED4_TR1 RED4_LED1 RED4_R1
RED3_TR1 RED3_LED1 RED3_R1
RED2_TR1 RED2_LED1 RED2_R1
RED1_TR1 RED1_LED1 RED1_R1

We need to placement and routing only one room!!!!

INTERACTIVE ROUTING - BASICS

To place a track, move the cursor to where you want the current section of track segments to end and **Left Click** or press **ENTER** - the track will be placed up to the current cursor position.

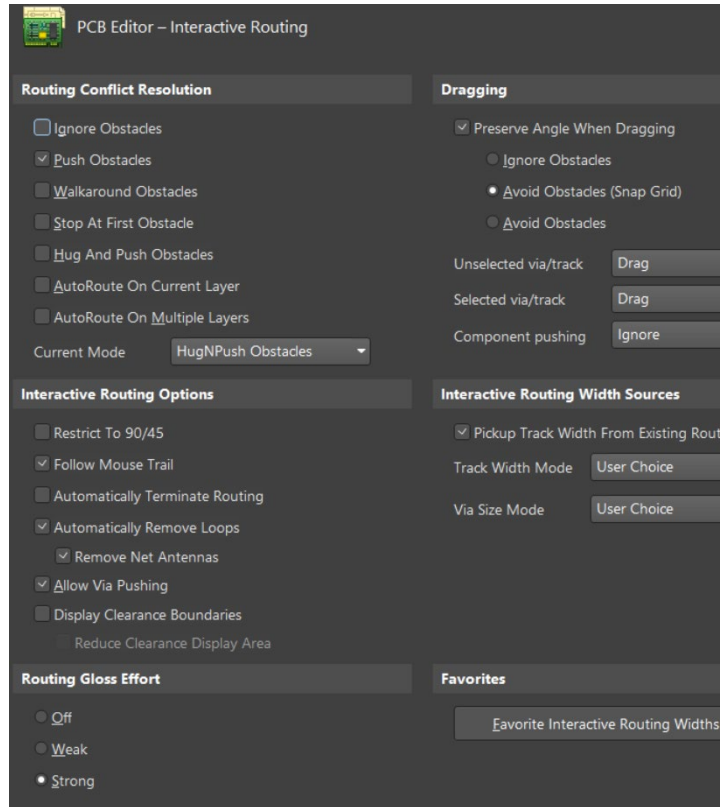
- **ESC** - Terminate the current route. Any routing that has been committed before calling the termination is retained.
- **BACKSPACE** - Unwinds the last committed route back to its starting point. If any objects had been *pushed* through placing the last segment, they are moved back to their original positions. This feature is not available after using Auto-Complete.
- **7** - Cycles through the connections available for routing if the current pad has multiple connections.
- **9** - Switches the cursor position from the currently selected pad or track to the target pad or track. If the location of the object being switched to is not in the current window, the view jumps and centers around the new cursor position.



LINE / TRACK

- click or press ENTER to anchor the starting point for the first track segment
- move the cursor to size the track and click or press ENTER to anchor the end point for this first segment, which is also the start point for the next connected segment
- position the cursor and click or press ENTER to anchor a series of vertex points that define the series of connected track segments
- right-click or press ESC to end the current series of connected track segments.
- Use **SHIFT** + SPACEBAR to cycle through various corner modes
 - Any angle, 45°, 45° with arc, 90° and 90° with arc.
- Use **SPACEBAR** to toggle the direction of the corner
- Continue placing further tracks, or right-click or press ESC to exit placement mode
- Use the BACKSPACE key while in placement mode, to remove the last placed track segment
- press the * key (the asterisk key on the numeric keypad) to cycle through the available signal layers
- press the + and - keys (on the numeric keypad) to cycle forward and backward through all visible layers

INTERACTIVE ROUTING



- **Controlling Corner Styles**

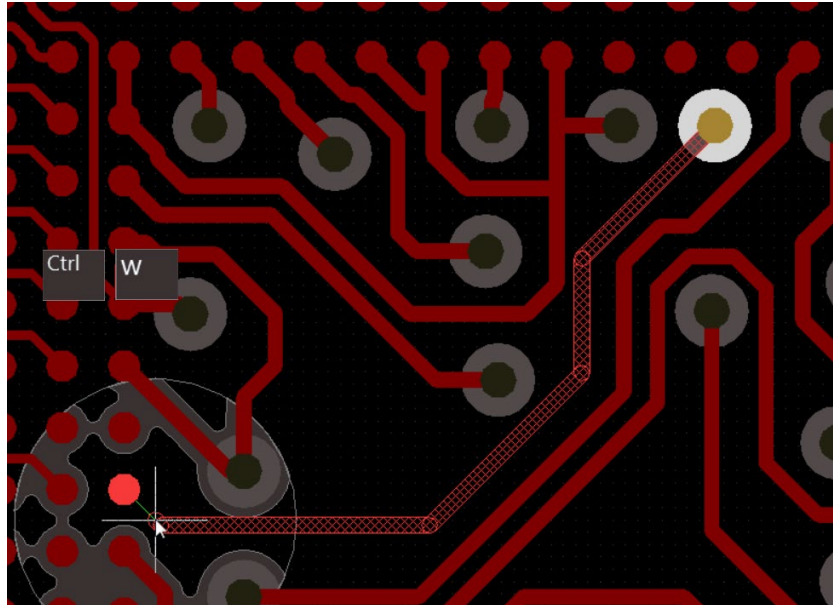
- Styles can be cycled through using **SHIFT + SPACEBAR**
- Any angle, 45°, 45° with arc, 90°, 90° with arc
- Arcs can be increased or decreased in radius using . (period or full stop) and , (comma) respectively.
- **SHIFT + .** and **SHIFT + ,** increase/decrease radius by a factor of 10.
- Use **SPACEBAR** to toggle the direction of the corner.

- **Automatic Connection Completion**

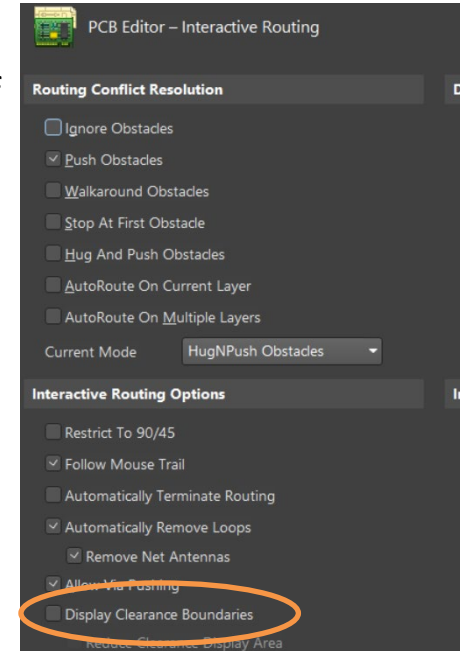
- The Interactive Router is able to attempt automatic completion (Auto-Complete) of connections to the target pad, hold **CTRL** and **Left Click** to instruct the Interactive Router to attempt to complete the current connection.
- Start point and target pad should be on the same layer
- The potential auto-completion route should be compatible to the existing design rules

INTERACTIVE ROUTING

Delivered as part of a set of improvements aimed at helping interpret and understand the impact of design rules, the new dynamic display of clearance boundaries feature will help you see just how much space is available during interactive routing.



CTRL + W



Enable the **Display Clearance Boundaries** feature in the **PCB Editor - Interactive Routing** page of the *Preferences* dialog.

INTERACTIVE ROUTING



- Listing shortcut keys

When a command is in use, Altium give us a list of different shortcut keys to increase the productivity. Could exists a long list of shortcut keys on a command, and these change between commands, these means an impossible task of remember all these shortcuts.

SHIFT + F1

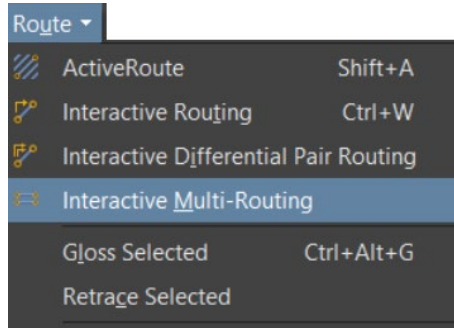
show us the complete list from the command selected.

Help	F1
Pause interactive process	Tab
Suspend	Esc
Commit	Enter
Undo Commit	BkSp
Autocomplete Segments To Target (Ctrl+Click)	
✓ Look Ahead Mode	1
Toggle Elbow Side	Space
Cycle Corner Style	Shift+Space
Toggle Routing Mode	Shift+R
✓ Toggle Follow Mouse Trail Mode	5
✓ Toggle Loop Removal	Shift+D
✓ Show Clearance Boundaries	Ctrl+W
Choose Favorite Width	Shift+W
Choose Favorite Via Size	Shift+V
Cycle Track-Width Source	3
Cycle Via-Size Source	4
Next Layer	TECLA DE ADICION
Next Layer	TECLA DE MULTIPLICACION
Previous Layer	TECLA DE SUSTRACCION
Switch Layer For Current Trace	L
Add Fanout Via and Suspend	ç
Add Via (No Layer Change)	2
Change routing Via Start/End layers	6
Next Routing Target	7
Swap To Opposite Route Point	9
Add Accordions	Shift+A
Toggle Length Gauge	Shift+G
Cycle Glossing Effort	Shift+Ctrl+G
Enable Subnet Swapping	Shift+C
Swap Target Subnet	Shift+T

INTERACTIVE ROUTING - VIAS

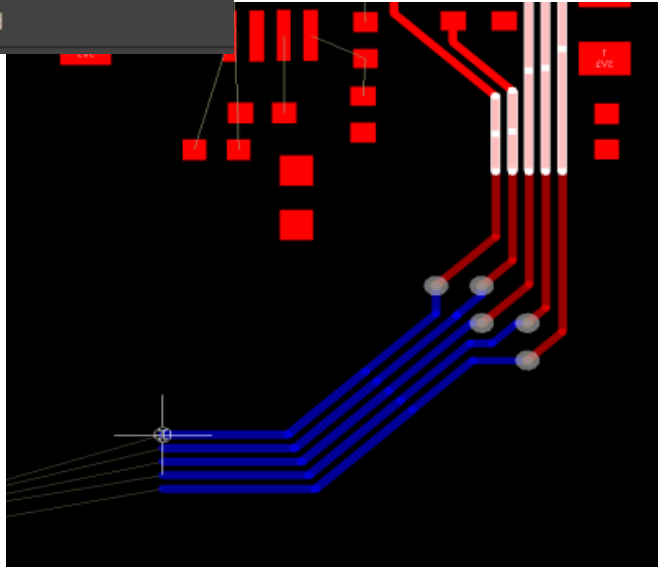
- Altium Designer provides you with ability to add vias on-the-fly during interactive routing.
 - Vias can be added in only valid locations
 - The software will prevent you from placing vias if they conflict with objects on any of the layers
 - The properties of the via are determined by the applicable Routing Via Style design rule in the *PCB Rules and Constraints Editor* dialog
 - Pressing 4 key, we can change the via models listed on rules.
- Adding a Via On Layer Change
 - Press the * (asterisk) or + (plus) key on the numeric keypad while routing to insert a via and switch the routing to the next signal layer.
 - Press the - (minus) key on the numeric keypad to insert a via and switch the routing to the previous signal layer..
- Adding a Via Without Layer Change
 - Press the 2 key while routing to insert a via, however, keep routing on the current signal layer. Click to commit the via position.

INTERACTIVE MULTI-ROUTING



This command is used to route multiple connections by placing track objects onto the current document and is useful for quickly routing connections manually.

To use this feature start by selecting pads or tracks to route from.



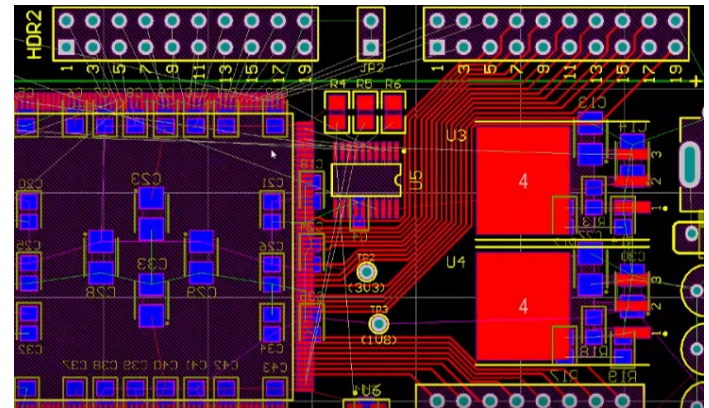
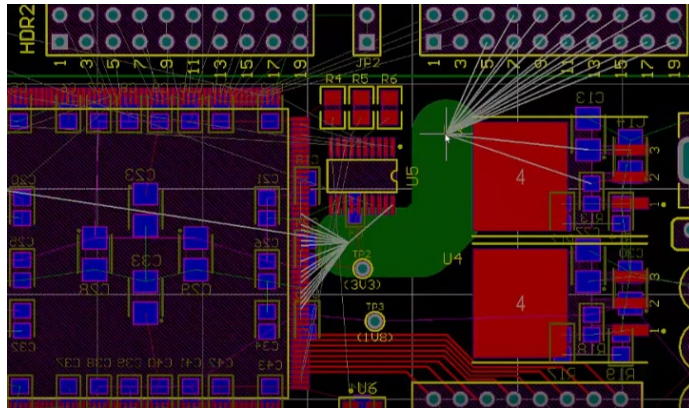
Similar to interactive routing, this tool helps maximize routing efficiency and flexibility shortcuts:

- C: converge bus spacing to rule.
- B: decrease bus spacing.
- SHIFT+B: increase bus spacing.

ACTIVE ROUTING

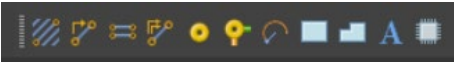
ActiveRoute is an automated interactive routing technology, that works with the designer to rapidly route selected connections. Rather than allowing an autorouter to do its best at routing the entire board, ActiveRoute acknowledges the reality that board design is a highly interactive process, where the best results are produced by skilled designers using powerful tools, under their control.

One of the great challenges with interactive software tools is interfacing those tools to the designer's fingertips, so they can easily and fluidly move from interactively routing through a tight area of the board, and then apply ActiveRoute's algorithms to route what remains.

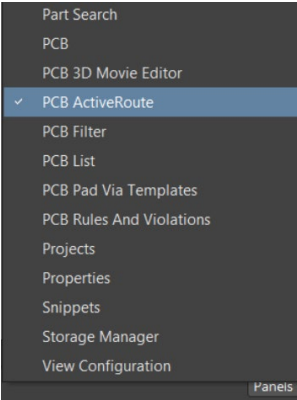
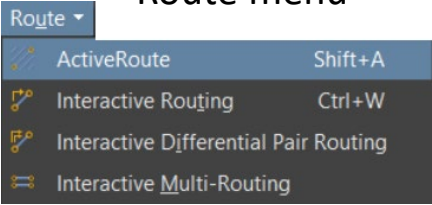


ACTIVE ROUTING

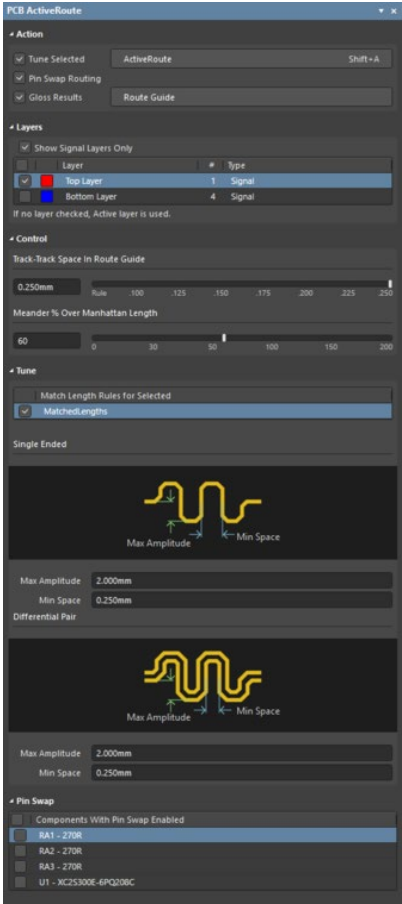
Wiring toolbox



Route menu



PCB panel list



ACTIVE ROUTING - SET UP

- Create net classes.
 - ActiveRoute requires the nets to be selected, having **net classes** can greatly assist in the selection process when there is a high number of nets to be ActiveRouted.
- Shelf existing polygons
 - ActiveRoute can route through existing polygons if the **Repour Polygons after Modification** option is enabled in the **PCB EDITOR - GENERAL** page of the *Preferences* dialog. Alternatively, polygons can be shelved (made invisible, but retained in the design), using the **Tools » Polygon Pours » Shelf Polygons** command.
- Configure the design rules.
- Fanout the design
 - ActiveRoute does not place vias, so you must create fanouts for BGAs, connectors, and discrete components that use SMT pins.
- Power the power and ground nets
 - ActiveRoute is for signal nets, it ignores nets assigned to power planes, and nets that contain more than 20 pins .

ACTIVE ROUTING - PERFORMING

Connections



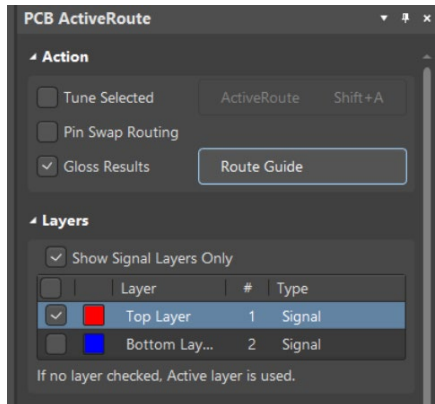
Existing routing



Component pads

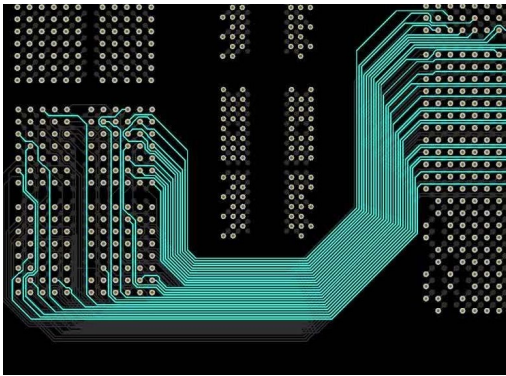
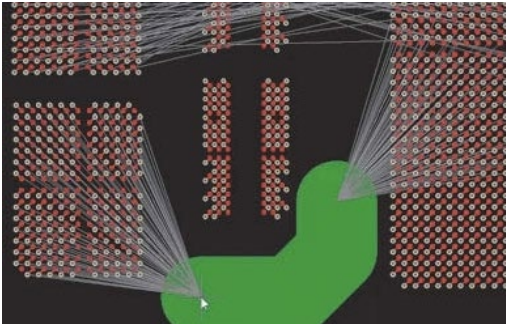


Other objects in the net



- Select the connections to be routed
 - ActiveRoute requires the nets to be selected, chose the different selecting techniques to obtain the required nets.
- Enable the layers to ActiveRoute
 - In the *PCB ActiveRoute* panel, enable the layer(s) that the selected connections are to be routed on. ActiveRoute will distribute the routes across these layers.
 - If no layer is selected, then it will route on the Active Layer, which is the selected tab at the bottom of the main PCB editor window.

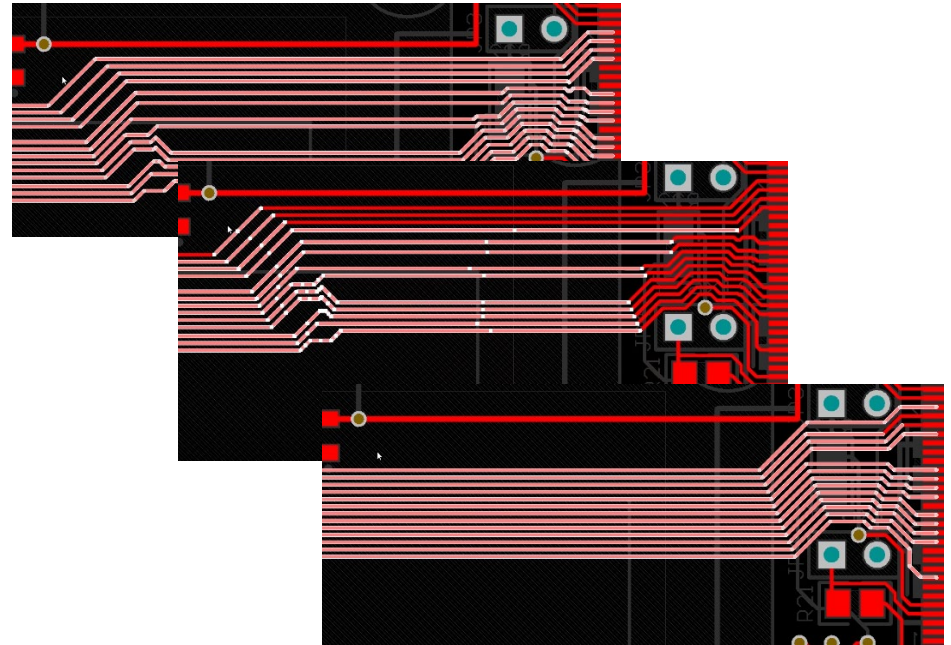
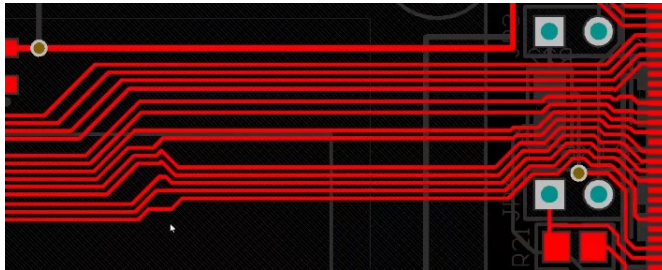
ACTIVE ROUTING - PERFORMING



- Define a Route Guide (if you want)
 - Click the **Route Guide** button in the *PCB ActiveRoute* panel.
 - The cursor will appear to be *holding* all of the selected connections, indicating that this set of connections must flow through the Guide. Move the cursor to a suitable start location, at either end of the path you wish to define.
 - Press the **Up Arrow** key to widen the Guide, or the **Down Arrow** key to narrow the Guide.
 - Continue this process of path definition, stopping some distance away from where the connection lines terminate, allowing room for the connections to flow out of the Guide to their target pads / vias / track ends.
- Running
 - Select the **Route » ActiveRoute** command, **ActiveRoute** button or use the **Shift+A** shortcut.
 - If you want to perform the connections you can Gloss the current selection **Ctrl+Alt+G**.

TRACK GLOSSING

Applied to selected tracks, the tool endeavours to improve the quality of the overall route defined by those track segments involved.

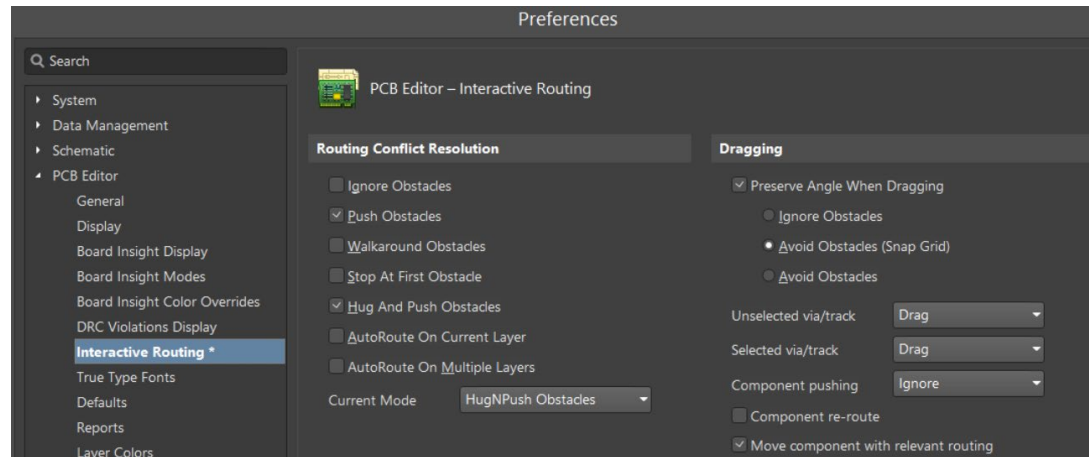


TRACK GLOSSING

- **Use:** ensure that the routed track that you wish to apply glossing to, is selected in the design workspace, *before* running the command.
 - Use **Click+Drag** a selection rectangle, from right to left, to invoke the select touching rectangle mode to quickly select a track segment in multiple nets.
 - Select a segment anywhere within the route, then press **Tab** to extend that selection.
 - To gloss only a section of a routed net, select a track segment at either end of the section, then run the **Gloss** command. Only the routing between (and including) the selected segments will be glossed.
- **Access:** the glossing tool can be accessed from the PCB Editor by
 - Choosing the **Edit » Gloss** (will become **Route » Gloss Selected**) command from the main menus.
 - Using the **Ctrl+Alt+G** keyboard shortcut.
- **Considerations**
 - Tracks at non-45 degree angle increments are not glossed, unless they are almost regular.
 - Arcs are currently not glossed.

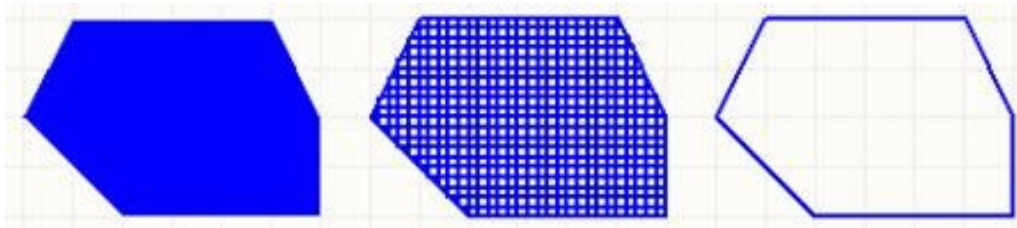
COMPONENT REROUTE

Component Re-route Mode (Shift+R) - toggles post-drop connectivity restoration. After the moving set has been released, the software will attempt to re-route the component(s) to reconnect any broken nets. Use the **Shift+R** shortcut to inhibit the re-route behaviour (or disable the **Component re-route** option in **Interactive Routing** page of the *Preferences* dialog)



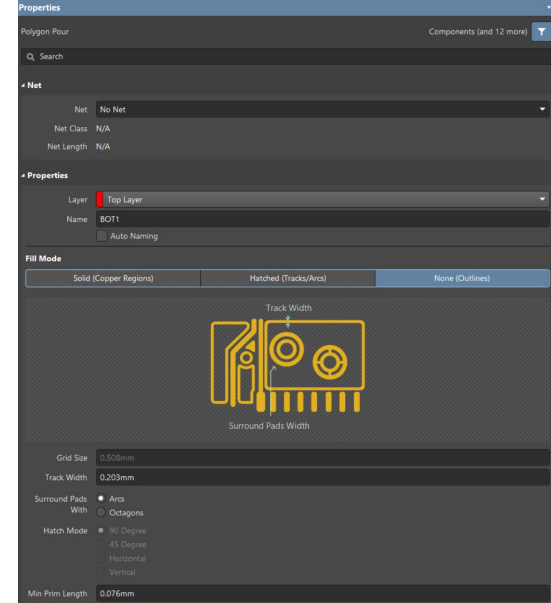
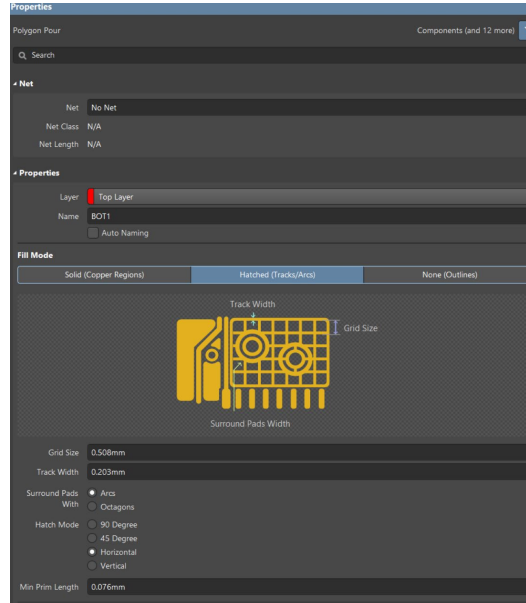
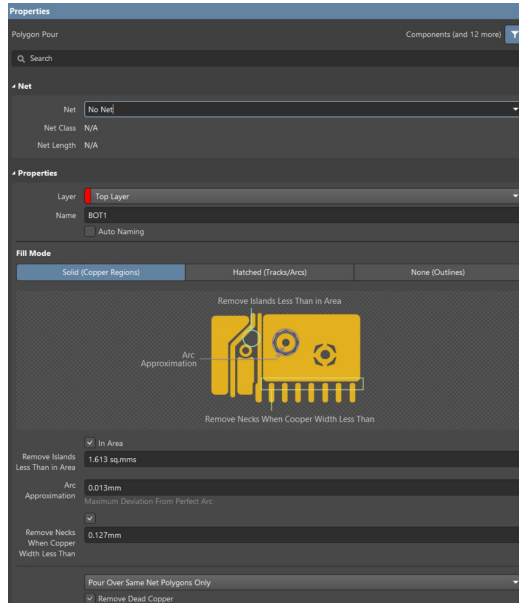
POLYGON POUR

- A polygon pour is a group design object
 - It creates a solid, hatch-filled (lattice) or outline-only area on the selected PCB layer
 - Also referred to as copper pours, they are similar to area fills, except that they can fill irregularly shaped areas of a board and can connect to a specified net as they are poured
- On a signal layer, you might place a solid polygon pour to define an area for carrying large power supply currents, or as a ground-connected area for providing electro-magnetic shielding
- On a non-signal layer, you might use solid or outline-only polygon pours to distinguish between specific functional areas of a board



POLYGON POUR TYPE

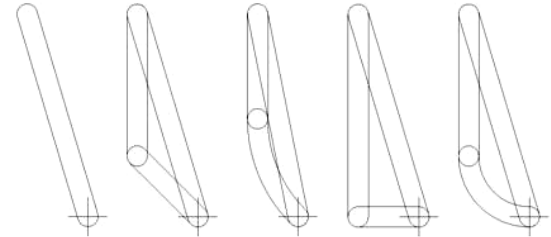
Solid, Hatched or None



CREATING POLYGON POURS

- Place ➡ Polygon Pour

- Position the cursor and click to anchor the starting point
- Position the cursor and click to anchor a series of vertex points
- Right-click or press ESC to complete placement of the polygon pour. The tool will automatically "close" the polygon
- While defining the shape of the polygon pour, use **SHIFT** + SPACEBAR to cycle through various corner modes.

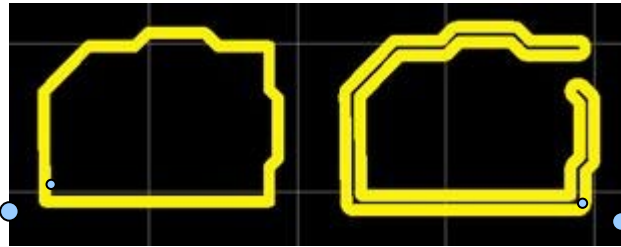


- Arcs can be increased or decreased in radius using **SHIFT** + . (period or full stop) or **SHIFT** + , (comma) respectively.
- Use **SPACEBAR** to toggle the direction of the corner.
- Use the **BACKSPACE** key while in placement mode, to remove the last placed vertex point. Repeatedly use this key to 'unwind' the outline for the polygon, right back to the initial starting point.

CREATING POLYGON POURS

- **Tools ➔ Convert ➔ Create Polygon from Selected Primitives**

- Create polygon objects in the form of polygon pours, regions, polygon cutouts or board cutouts



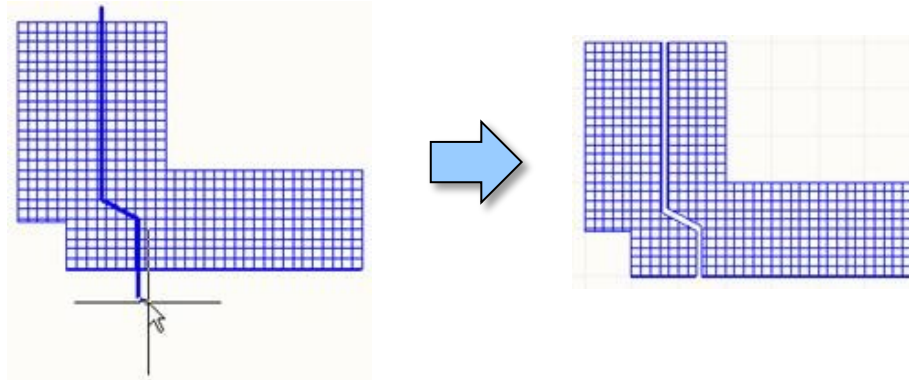
Polygon from
closed outline

Polygon from
open outline

SLICING POLYGON POURS

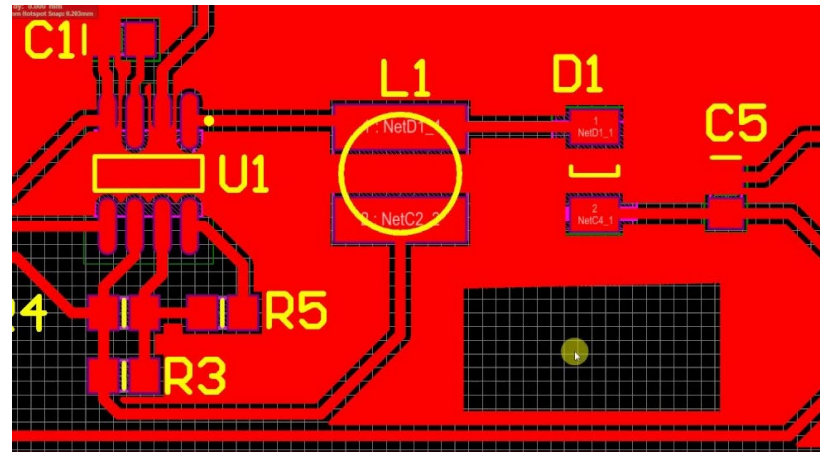
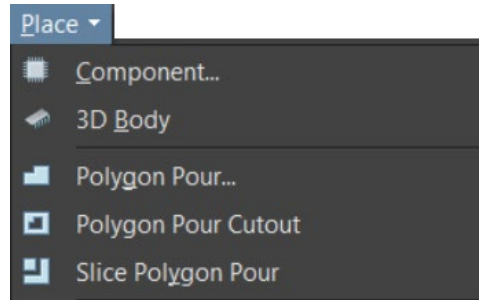
A placed polygon pour object can be graphically 'sliced' into two or more separate polygon pours.

- **Place ➔ Slice Polygon Pour**
- After launching the command, filtering will be applied to the document, resulting in all polygon pour objects on the current layer remaining visible and all other objects becoming dimmed



POLYGON POUR CUTOUT

- There is an additional command in the place menu:
 - Polygon Pour Cutout
- It simply adds a region with the polygon cutout attribute:

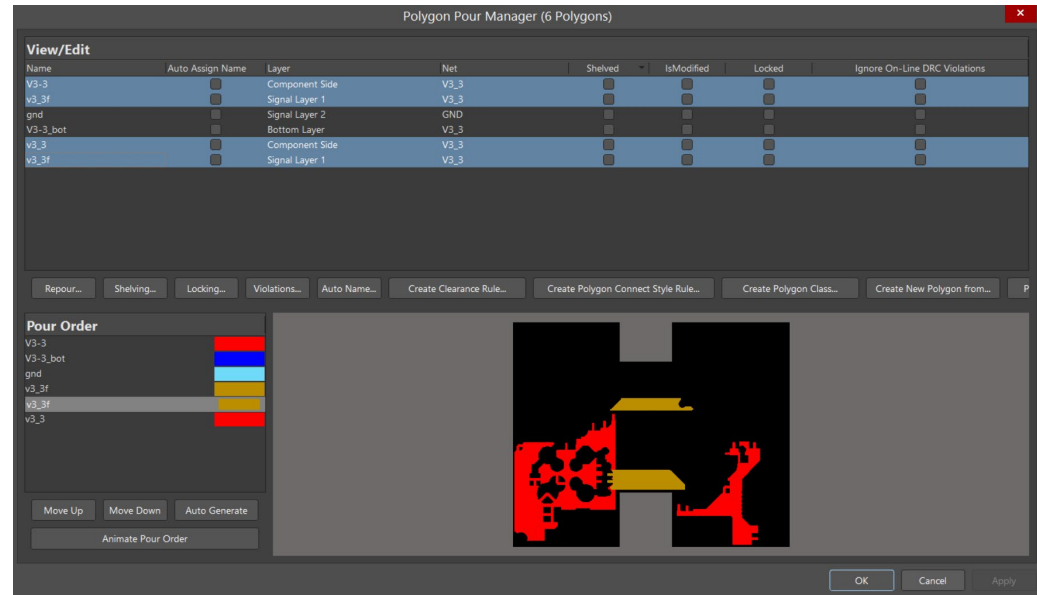
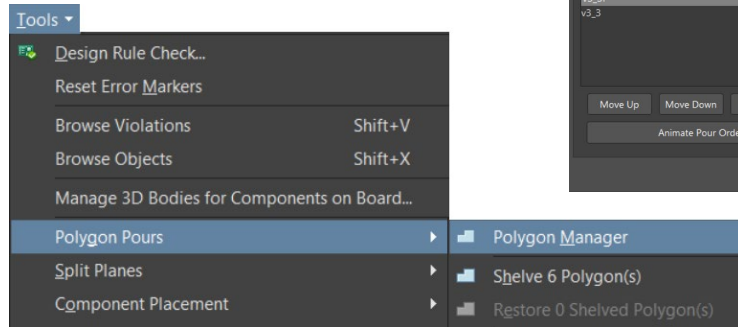


OPERATIONS ON POLYGON POURS

- Converting Hatched Polygon Pours
 - Multiple hatched polygon pours may be converted to solid polygon pours, simultaneously, using the Tools » Polygon Pours » Convert Hatched Polygons To Solid
- Shelving Polygon Pours
 - If a design has numerous polygon pours, especially of the larger, hatch-filled variety, the repouring phase can be quite slow.
 - To alleviate this, you could temporarily change the Fill Mode of all such pours to None, meaning that only the outline tracks will be displayed.
 - Alternatively, a more convenient method is hide all polygon pours, which is known as Shelving.
- To shelve all polygons in the current design, select the Tools ➡ Polygon Pours ➡ Shelf Polygon(s)
- To reinstate polygons that have previously been shelved, select the Tools ➡ Polygon Pours ➡ Restore n Shelved Polygon(s)

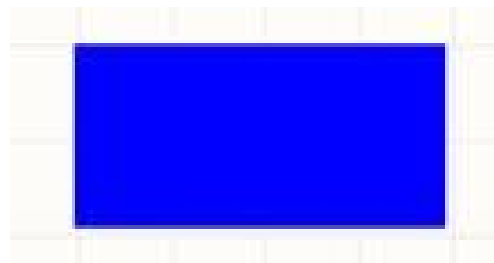
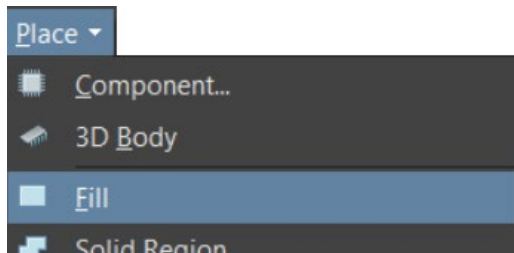
POLYGON POUR MANAGER

The *Polygon Pour Manager* dialog provides a high-level view of all polygons on the entire board



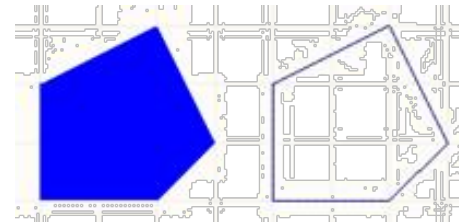
FILL

- A fill is a primitive design object.
 - It is a rectangular object that can be placed on any layer.
 - When placed on a signal layer, a fill becomes an area of solid copper that can be used to provide shielding or to carry large currents.
 - Fills of varying size can be combined to cover irregularly shaped areas and can also be combined with track or arc segments and be connected to a net.
 - Fills can also be placed on non-electrical layers. For example, place a fill on the Keep-Out layer to designate a 'no-go' area for both autorouting and autoplacement.
 - Place a fill on a Power Plane, Solder Mask, or Paste Mask layer, to create a void on that layer (on negative layers)



SOLID REGION

- A region is a primitive, polygon-type object that can be placed on any layer.
 - It can be configured to be positive (placed as a copper region) or negative (placed as a polygon pour cutout) or be multi-layer (placed as a board cutout)
- When using it as a positive region, it is similar in nature to its rectangular-based fill counterpart.
 - When placed on a signal layer a positive region becomes an area of solid copper that can be used to provide shielding or to carry large currents.
 - Positive regions can be combined with track or arc segments and be connected to a net.
 - Positive regions can also be placed on non-electrical layers. For example, place a region on the Keep-Out layer to designate a 'no-go' area for both autorouting and autoplacement.
 - Place a region on a Power Plane, Solder Mask, or Paste Mask layer, to create a void on that layer (negative layers)



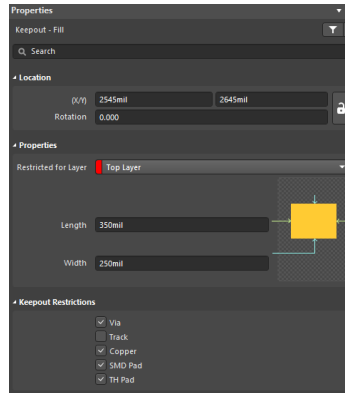
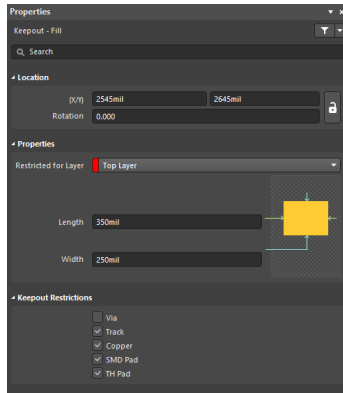
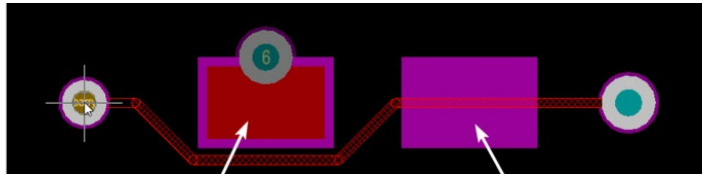
SOLID REGION

- When used as a negative copper region (polygon pour cutout) it provides a polygonal void area for use within solid polygon pours.
 - Such a region/cutout will not be filled with copper when the pour is flooded.
- When used as a board cutout it defines areas that are actual holes through the PCB and are independent of board layers.
 - Board cutout regions are transferred to Gerber and ODB++ files for manufacturing purposes
 - Regardless of which layer you are on when creating a board cutout, it will remain multi-layer



KEEPOUT ZONE

A **Keepout** in PCB design is a user defined area or perimeter placed in the layout that copper objects cannot intersect. Typically included to control the area used by automated copper placement actions, such as polygon pours and interactive routing, a Keepout also represents an invalid location when manually placing copper objects.



A Keepout is placed in the editor workspace from the **Place » Keepout** menu. Select the Keepout's Properties to edit its physical characteristics, Layer or object type restrictions.

The related *Keepout* dialog options are:

Restricted for Layer – sets the Keepout Layer, and therefore the board layer on which copper objects will be restricted (kept out).

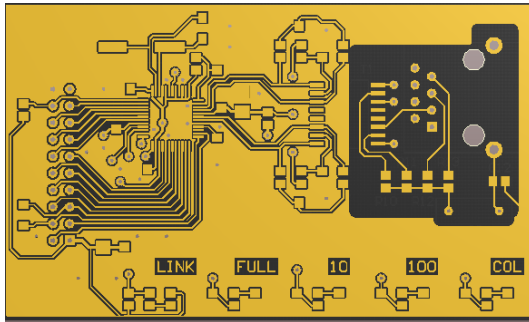
Keepout Restrictions – determines which object types will be restricted by the Keepout. Deselecting an object type will cause the Keepout to allow transgressions by that type of object (not kept out).

KEEPOUT ZONE

The inherent flexibility of Keepouts allows their use for a wide range of tasks to control PCB layout:

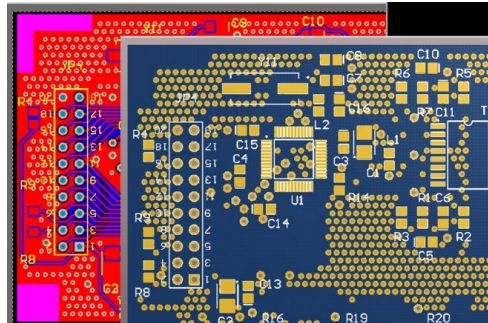
Polygon Pour control

The Keepout Fills are set to All Layers and configured to restrict only Copper objects, which will reject Polygon Pours, Fills and Regions while accepting existing tracks and pads etc



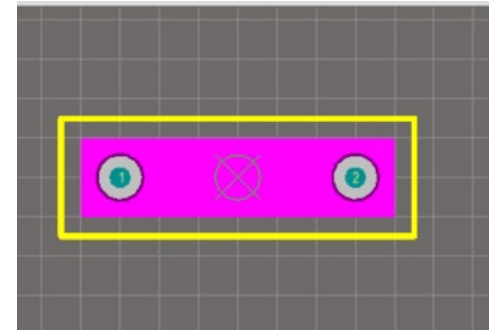
Via Stitching control

The application of Keepout shapes, configured to restrict Via objects, on multilayer copper areas can control the extent of automated Via Stitching

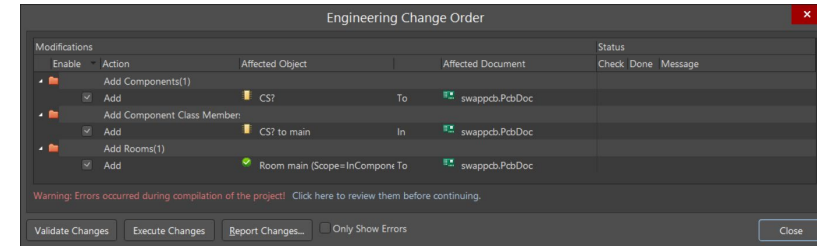
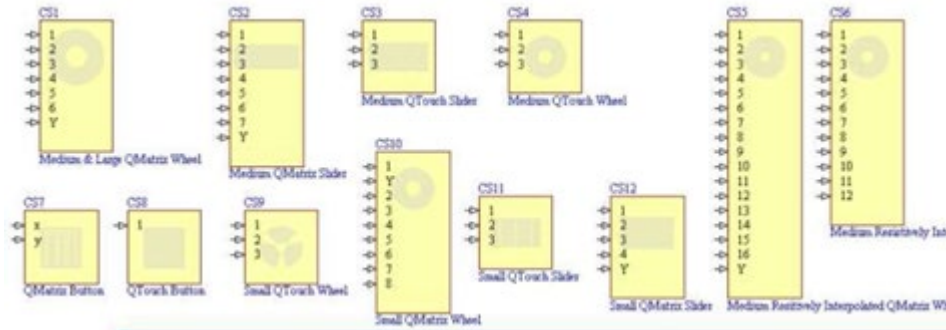


Keepouts in Library Components

Keepout are added to component Footprints in the PCB Library Editor using the same approach as those applied within the PCB Editor workspace.



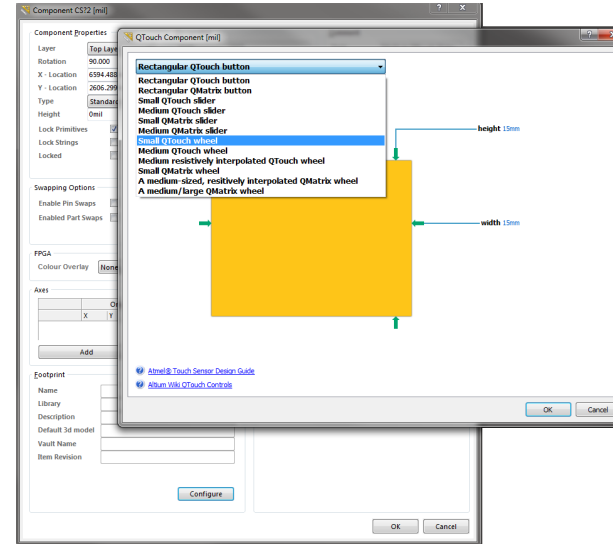
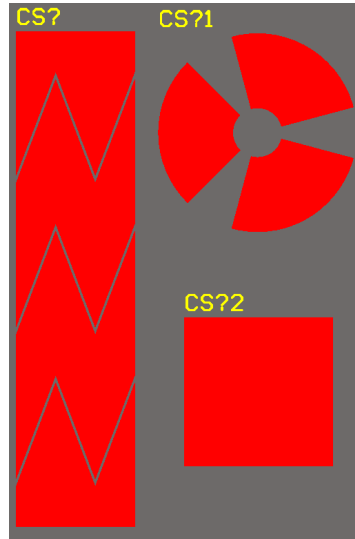
TOUCH CONTROL



A touch sensor is implemented in a design by placing and configuring the required sensor type from the dedicated Atmel Qtouch integrated library (Atmel QTouch.IntLib), located in the \Library\QTouch folder of the installation.

When configured as required, simply update the target PCB – an ECO is used to effect the required changes, resulting in the creation of the sensor pattern for placement on the PCB. Each sensor component on the PCB isn't a footprint in the normal sense, but rather the actual copper electrode pattern.

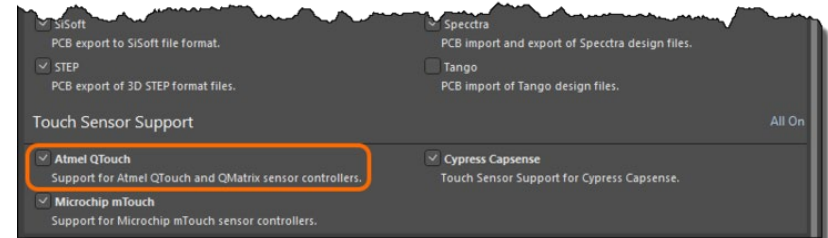
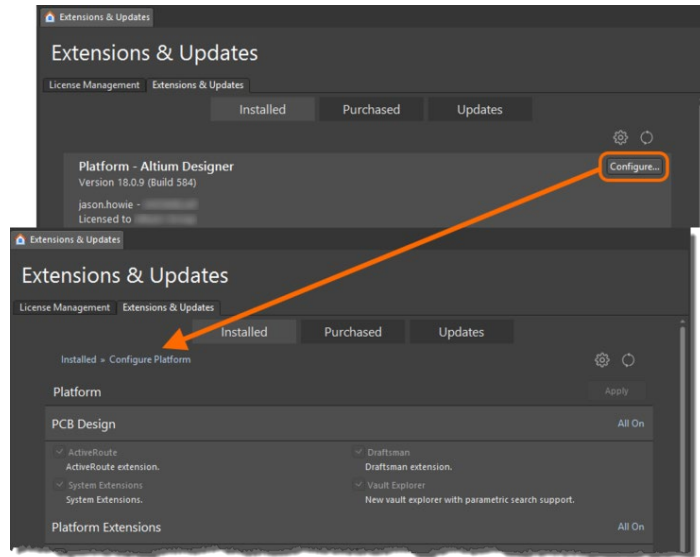
TOUCH CONTROL



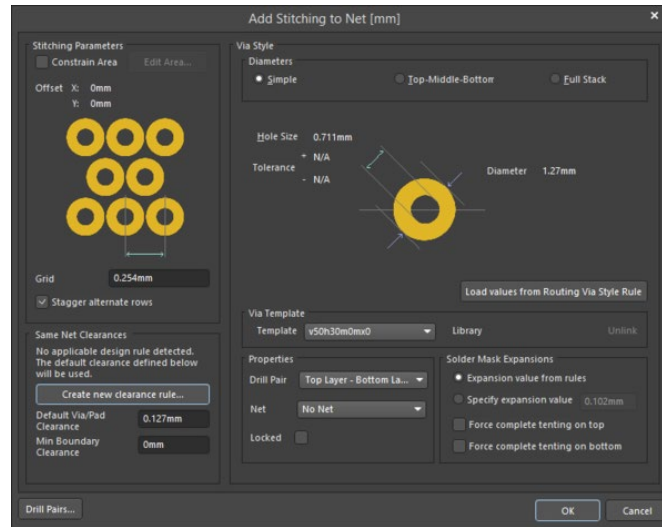
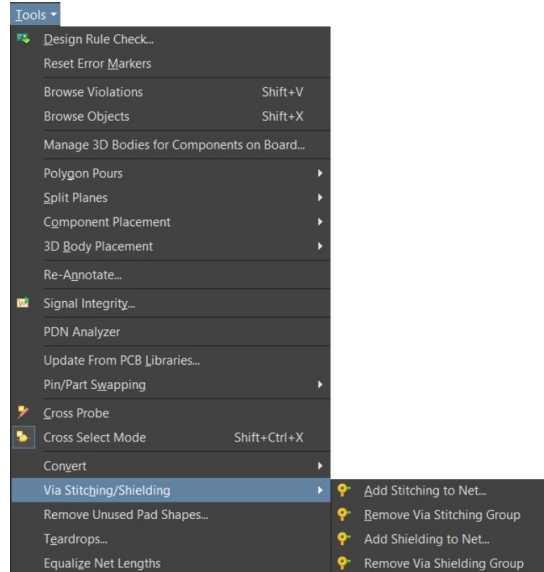
As with all configurable schematic components in Altium Designer, access to the associated configuration dialog for a sensor component is made by right-clicking over the component and choosing **Configure** from the context menu. Alternatively, click the **Configure** button from the associated properties dialog for the component.

TOUCH CONTROL.

If support has not already been added during initial installation of the software, it can be added from the **Configure Platform** page



VIA STITCHING

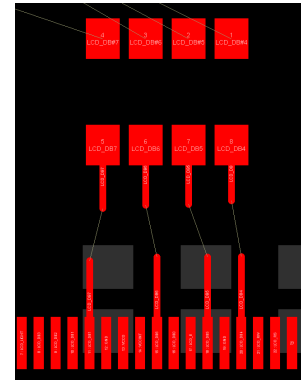
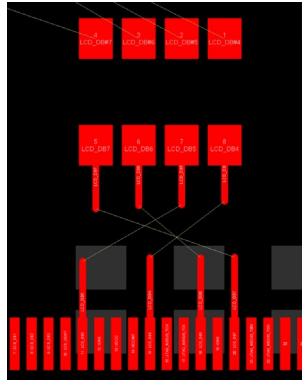


Via stitching is a technique used to tie together larger copper areas on different layers, in effect creating a strong vertical connection through the board structure, helping maintain a low impedance and short return loops.

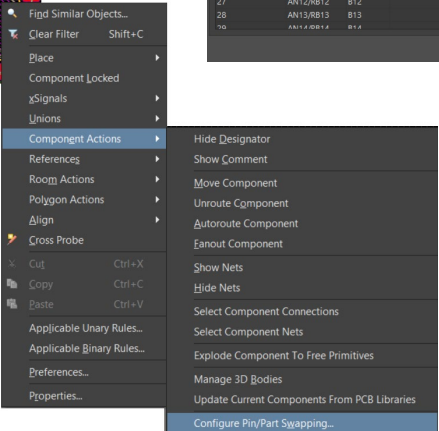
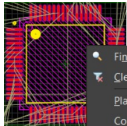
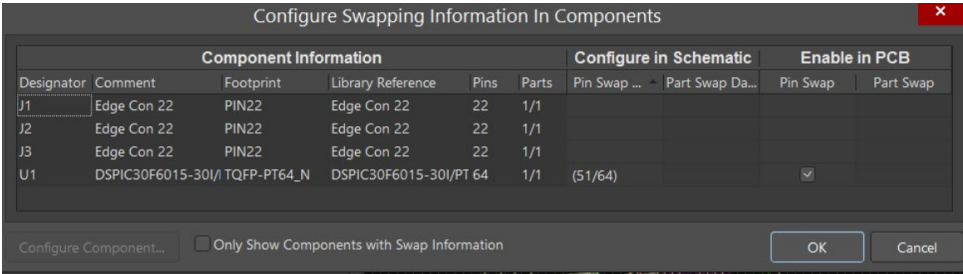
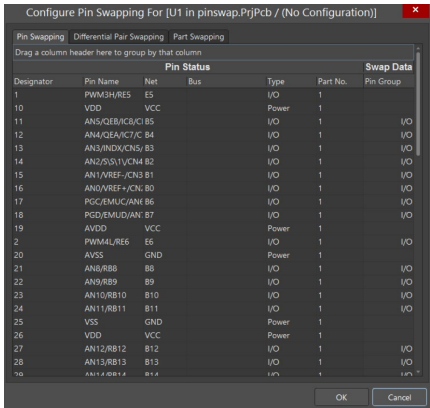
PIN\PART SWAPPING

This feature provides all the benefits of traditional pin-swapping systems. During a pin swap operation Altium Designer analyses the net assigned to the chosen pin, and dynamically reassigns the net on the pin and any connected copper.

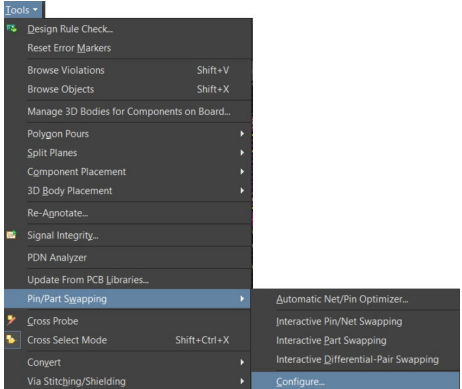
At the PCB level the system includes a powerful automatic optimizer that uses this information to dynamically re-assign nets to improve routability.



PIN\PART SWAPPING

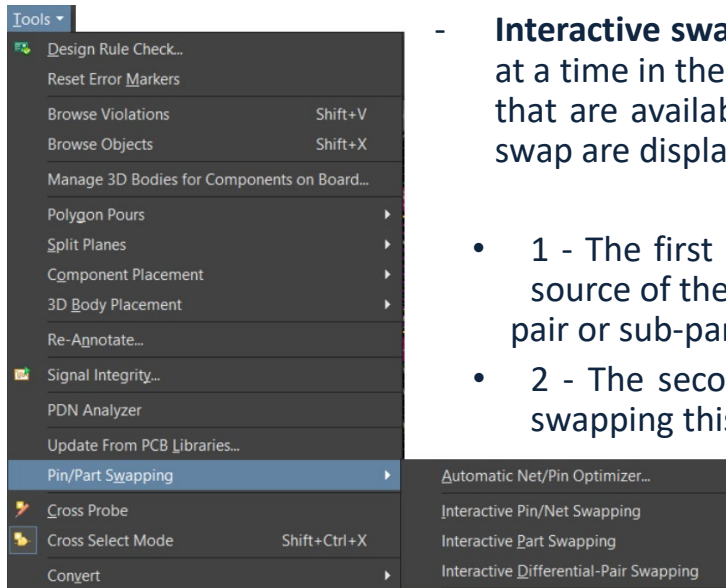


The Swap Manager lists all components used in the design, with their current swap settings. The PCB editor Swap Manager includes additional columns for enabling/disabling swapping on each component on the board.



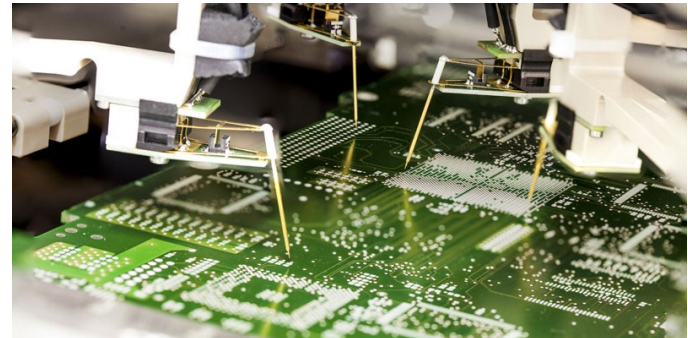
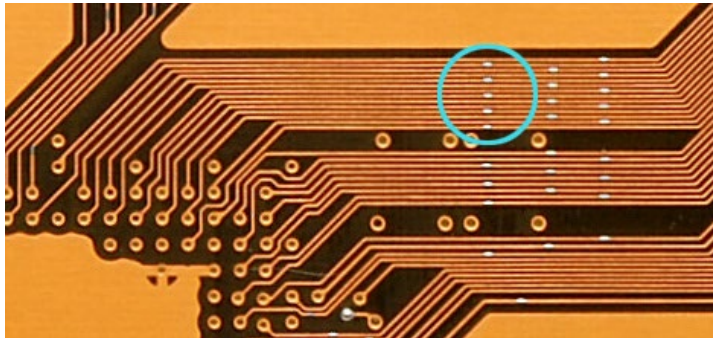
PIN\PART SWAPPING

- **The Automatic Pin/Net Optimizer** is a two-stage tool, that first runs a fast single-pass optimizer that attempts to minimize cross overs and connection lengths, but may actually increase them. When this is complete you will be asked if you want to run the iterative optimizer. The iterative optimizer will perform multiple passes in an attempt to reduce the number of cross overs and connection lengths.
- **Interactive swapping** allow pins, differential pairs or subparts to be swapped one at a time in the PCB editor. Once the command is selected from the menu the pins that are available for swapping are highlighted. The steps required to perform a swap are displayed on the Status line;
 - 1 - The first step is to select one of the highlighted pins that will become the source of the pin swap. In the case of pair or part swapping the differential pair or sub-part that the pin belongs to will be subsequently swapped.
 - 2 - The second step is to select the target pin for the swap. For pair or part swapping this pin will be representative of a differential pair or a sub-part.



TEST POINTS

Testing is an important part of the board manufacturing process. After fabrication, the board is typically tested to ensure no short or open circuits. Once fully populated with all its components, a board is often tested again to ensure signal integrity and device operation. To aide in this process, it is fundamentally beneficial to have a scheme of points on the board which the testing equipment can probe and perform the required tests.

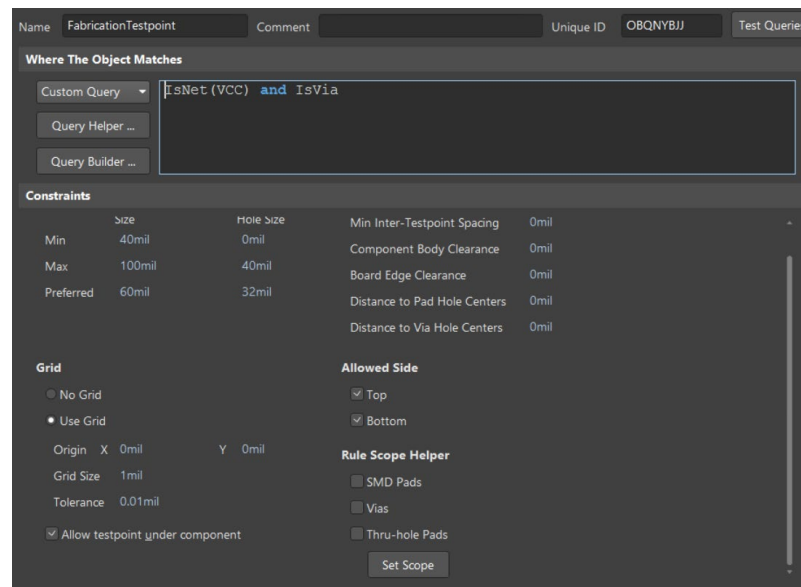


TEST POINTS

The constraints of a PCB design should be thought out and implemented as a well-honed set of design rules. To implement a successful testpoint scheme, where all defined testpoints can be accessed and used as part of the bare-board and/or in-circuit testing, governing constraints must be put in place. To this end, the following rule types are definable as part of the PCB Editor's Design Rules system:

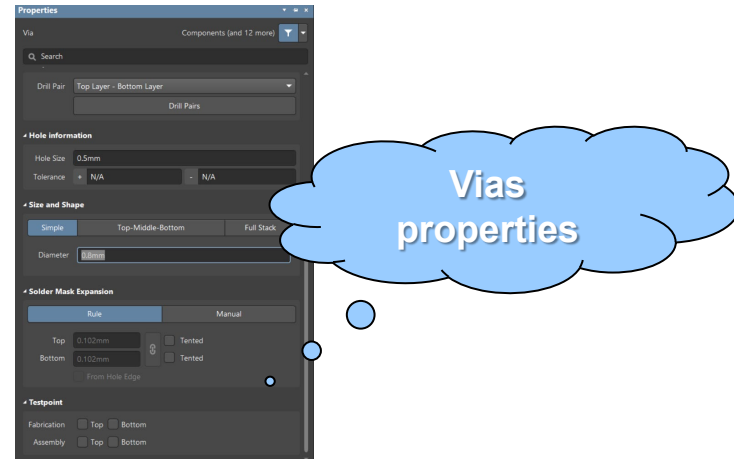
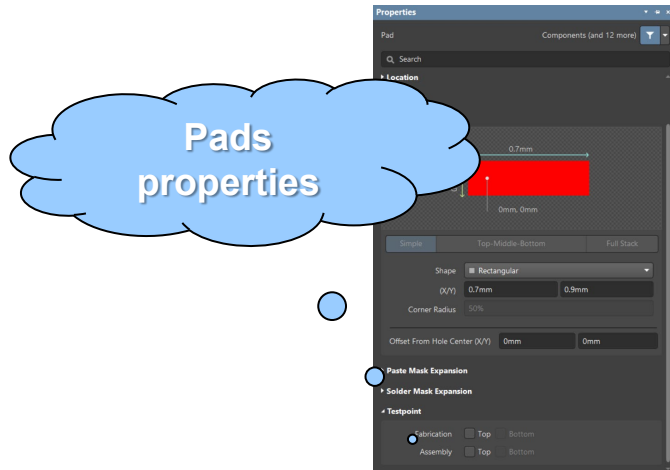
- Fabrication Testpoint Style
- Fabrication Testpoint Usage
- Assembly Testpoint Style
- Assembly Testpoint Usage

The style rule essentially specifies constraints that a pad or via has to meet in order to be considered for selection as a testpoint location. The usage rule simply specifies which nets require a testpoint.

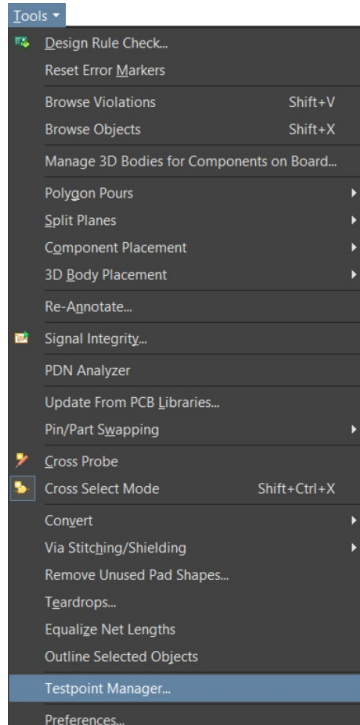


TEST POINTS

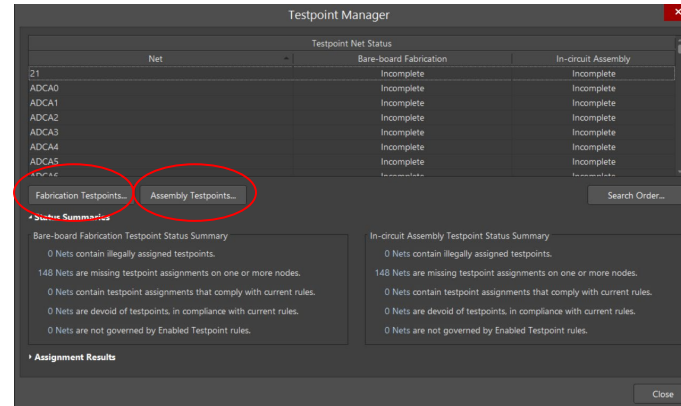
Altium Designer provides full support for testpoints, allowing you to specify pads (thru-hole or SMD) and/or vias to be used as testpoint locations in fabrication and/or assembly testing. A Pad or Via is nominated for use as a testpoint by setting its relevant testpoint properties – should it be a fabrication or assembly testpoint, and on which side of the board should it be used as a testpoint. These properties can be found in the *Pad* or *Via* properties dialogs.



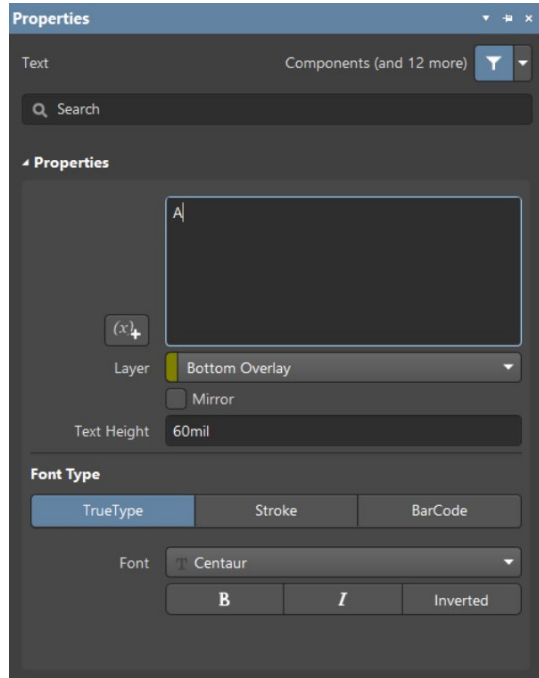
TEST POINTS



Assigning testpoints manually can be a laborious job at the best of times. Imagine this task on a more complex board, populated with hundreds of components and the process cries out for a more automated method of testpoint assignment. To cater for streamlined management of testpoints in your board designs, Altium Designer equips the PCB Editor with a **Testpoint Manager**. This *Testpoint Manager* search for you for existing Vias and Pads that can be used as testpoints.



WORKING WITH LOGOS



- The best way (and unknown) to insert logos and images to our PCB is using the window Font's system.
- Easy way to insert pre-saved logos or images.
- The fastest re-dimension and rotation way.



EMBEDDED BOARD ARRAY

An embedded board array is a primitive design object.

- It allows you to create a PCB panel (representing the physical board that the PCB is to be manufactured from) as part of your PCB design project.
- This is also known as *panelization*

